电路础
期末复习

针对课程：EE104/SME103　电路基础

自我介绍

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欢迎同学们加入深港微电子学院！

Content 期末复习内容

1. 直流电路分析
2. Sinusoids and Phasors 向量分析
3. Steady-state and power analysis 稳态电路和功率分析
4. Frequency response 频域响应

（第二节课）

5. Magnetically coupled circuits 磁耦合电路
6. Laplace Transform拉普拉斯变换
7. fx9750 使用教程

Review

电路基础知识结构

直流电路

基本电路定律（1-4、6）
运算放大器（5）
一阶RC、RL电路（7）
二阶RLC电路（8）

交流电路

交流电路定律（9-10）
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三相电路（12）
磁耦合电路（13）
频率响应（14）

高级电路分析

拉普拉斯变换（15-16）或
傅里叶变换（17-18）

Review

时域分析

以时间作为自变量的分析方式

频域分析

以角频率作为自变量的分析方式；傅里叶变换

S域分析

拉普拉斯变换

分析角度

直流电路分析

1-5

Chapter

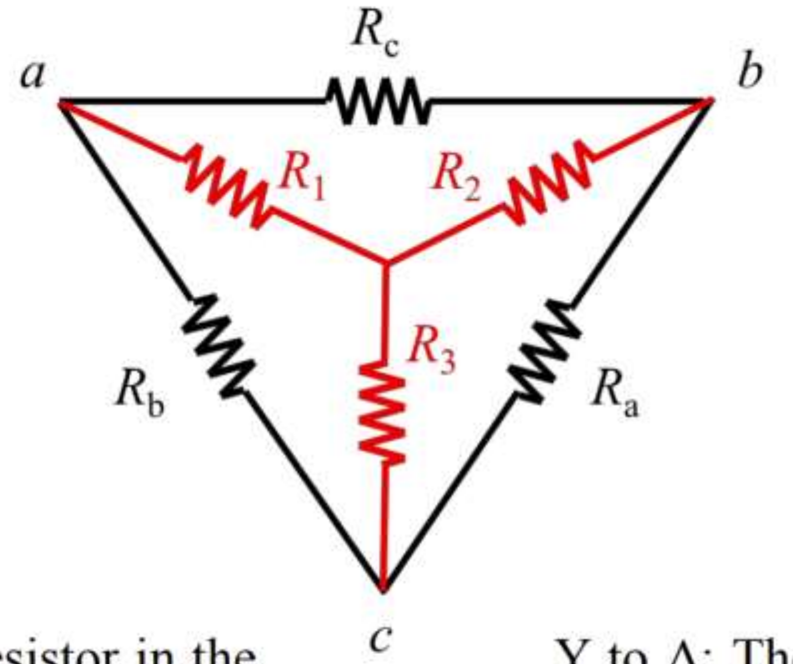
2.4 Δ-Y 转换

Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

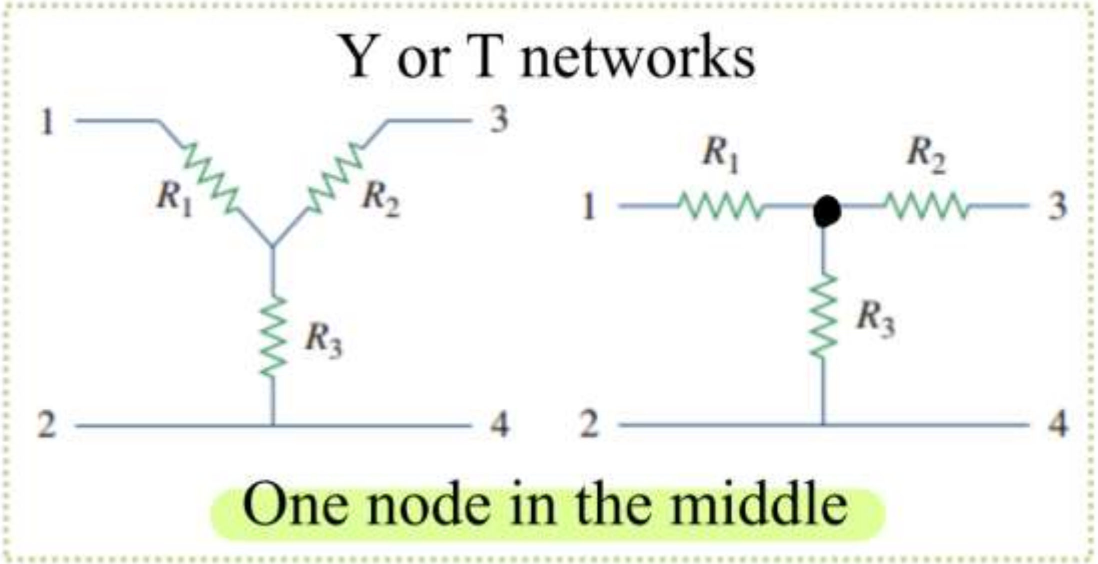
Δ to Y: the resistance of a resistor in the Y network is the product of the resistance of the two adjacent resistors in the Δ network, divided by the sum of the resistance of the three Δ resistors.

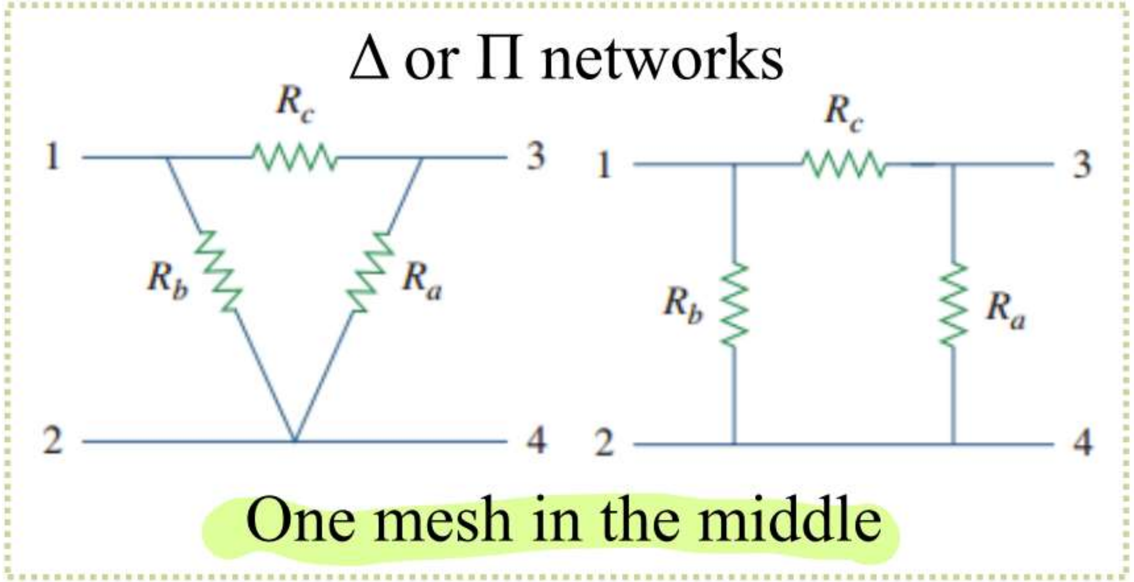
Y to Δ: The resistance of a resistor in the Δ network is the sum of all possible products the resistance of the Y resistors taken two at a time, divided by the resistance of the opposite Y resistor.

2.4 Δ-Y 转换

常见Y电路：

常见Δ电路：





转换的目的：化简电路，减少支路

源变换YYDS!

电路化简哪家强？

减少网孔数量　　网孔电流法

减少节点数量　　节点电压法

方程未知数数量减少

下笔前先观察是否能源变换

4.3 戴维南定理和诺顿定理

戴维南定理

VTh：端口开路电压

RTh：独立源关闭时端口的输入电阻

求RTh:

（1）当网络中不包含受控源时，关闭所有独立源，R_{Th}就是从ab两端向网络看进去的输入电阻

（2）当网络中包含受控源时，关闭独立源，

选择一下两种方法任意一种：

1.在ab端外加一个电压源V_0,计算端口电流i_0

2.在ab端外加一个电流源i_0，计算端口电压V_0

Norton’s Theorem诺顿定理

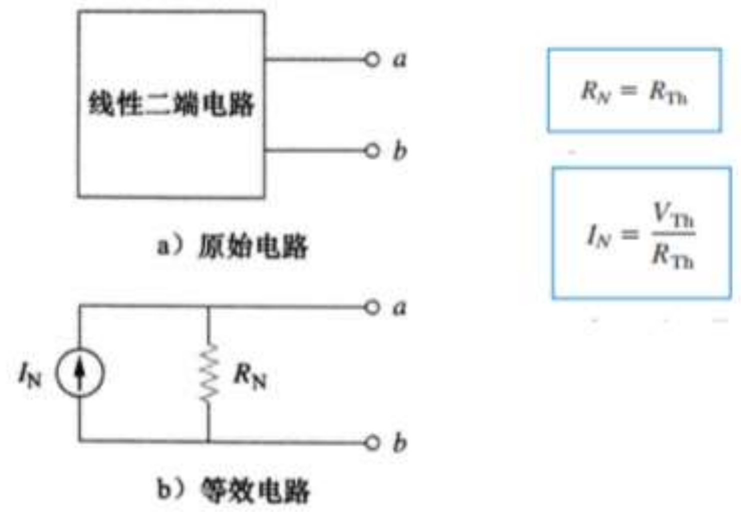
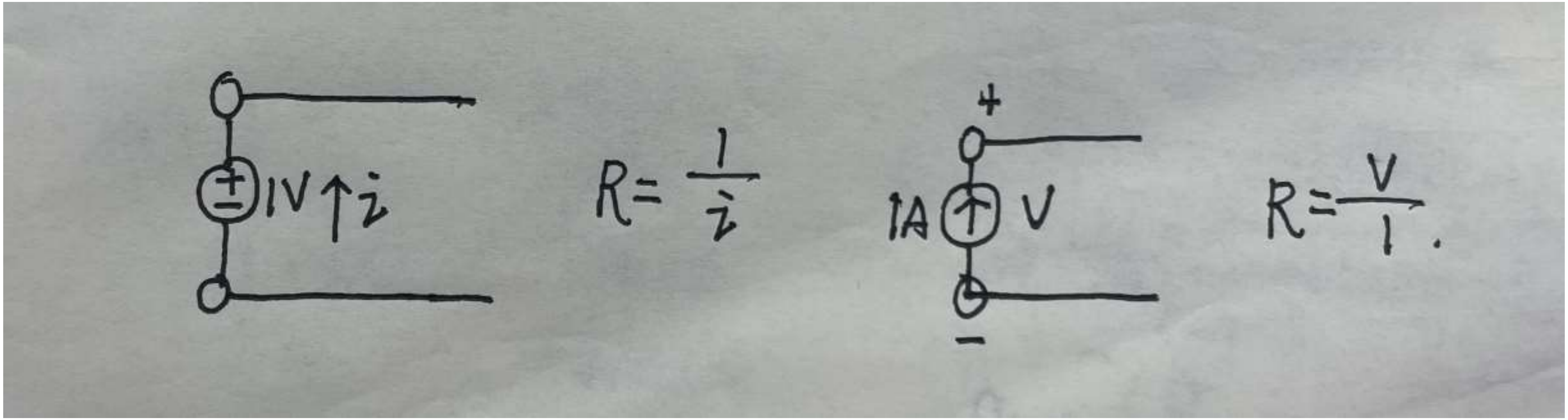
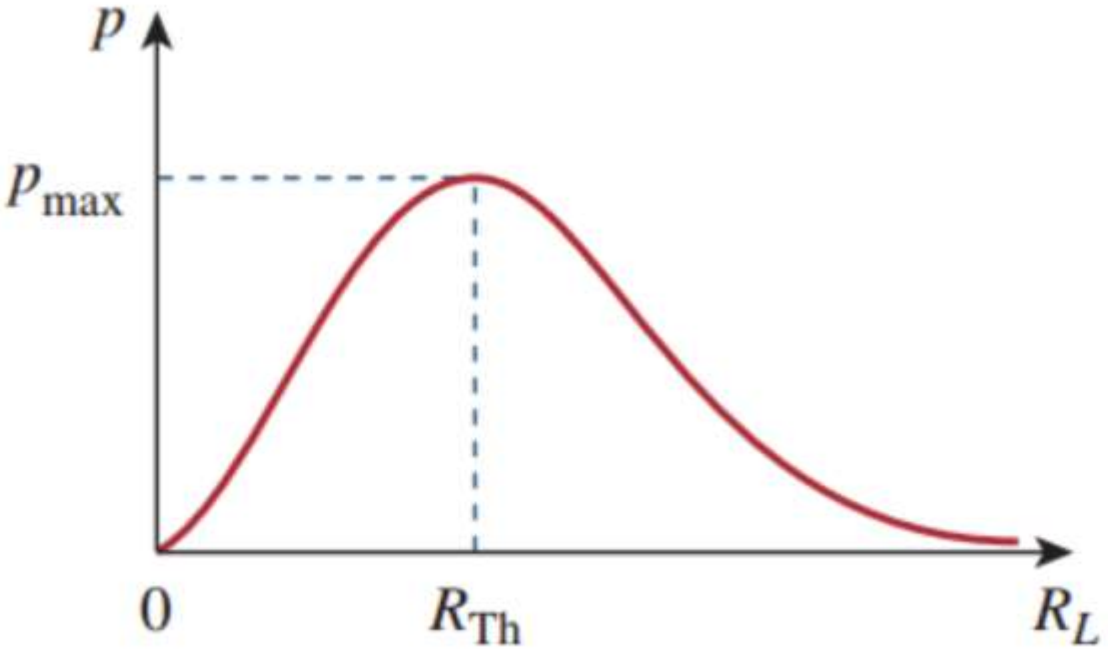


图 4-37 诺顿等效电路

易错提示：求等效电阻时的方向问题



4.4 最大功率传输定理



$$R_L = R_{\mathrm{Th}} \qquad p_{\max} = \frac{V_{\mathrm{Th}}^2}{4R_{\mathrm{Th}}}$$

$\boldsymbol{R_{th}}$：从负载电阻看进去的**戴维南等效电阻**

电容，电感和一阶，二阶电路

6-8

Chapter

6.1 电容（ Capacitors ）

几个重要的关系：

$$\boxed{C = \frac{\varepsilon A}{d}}$$

$$\boxed{q = Cv}$$

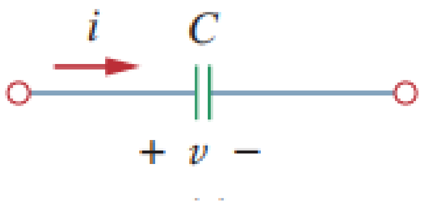
$$\boxed{i = C\frac{dv}{dt}}$$

$$\boxed{v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)}$$

$$W = \frac{1}{2}C\,V^2 = q^2/2C$$

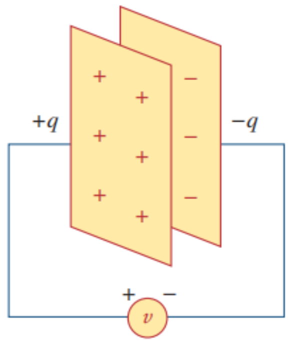
并联：

$$C_{eq} = C_1 + C_2 \ldots\ldots + C_n$$



串联：

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \ldots\ldots + \frac{1}{C_n}$$



对于直流电路：开路

电容上电压不会突变

6.2电感（ Inductors）

几个重要的关系：

$$L = \frac{N^2 \mu A}{l}$$

$$v = L\frac{di}{dt}$$

$$p = vi = \left(L\frac{di}{dt}\right)i = iL\frac{di}{dt}$$

$$i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)d\tau = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$$

$$W = \frac{1}{2}Li^2$$

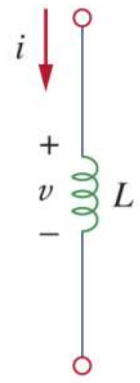
串联：

$$\mathbf{L}_{eq} = \mathbf{L}_1 + L_2 \ldots\ldots + L_n$$

并联：

$$\frac{1}{L_{eq}} = \frac{1}{L1} + \frac{1}{L_2} + \frac{1}{L_3} \ldots\ldots + \frac{1}{L_n}$$

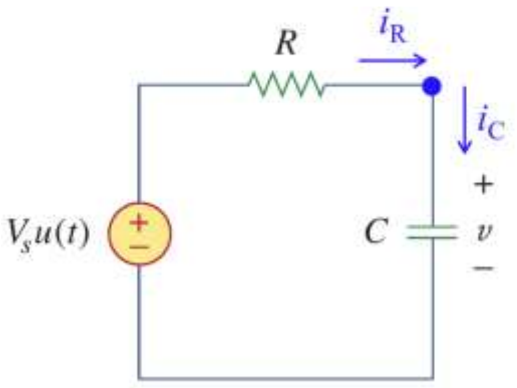
对于直流电路是导线
其电流不能突变



Air-core inductors

7.2阶跃响应（ Step-response of RC and RL circuits ）

阶跃响应分析方法：



1. 自由响应+受迫响应 （ forced response）

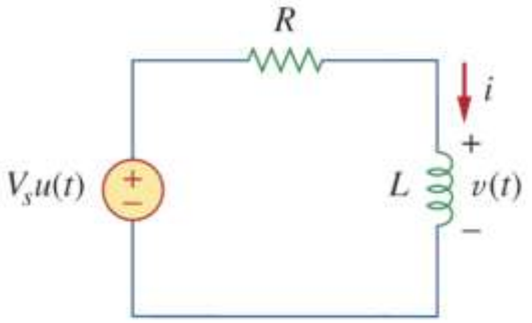
$$v(t) = v_n + v_f = v(0)e^{-t/\tau} + v(\infty)[1 - e^{-t/\tau}]$$

$$i(t) = i_n + i_f = i(0)e^{-t/\tau} + i(\infty)[1 - e^{-t/\tau}]$$

2. 暂态响应+稳态响应(transient response + steady-state response)

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau}$$

$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)]\, e^{-t/\tau}$$



都学过拉普拉斯变换了，还需要背记二阶电路的公式吗？

很有必要！如果题目考二阶电路，直接套公式速度远快于拉普拉斯变换！

Applying KVL around the loop, we have

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau = 0 \longrightarrow \boxed{\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0}$$

– This is a homogeneous second-order DE. Its initial conditions are:

$$\boxed{\begin{cases} i(0) = I_0 \\ \dfrac{di(0)}{dt} = \dfrac{0 - RI_0 - V_0}{L} = -\dfrac{1}{L}(RI_0 + V_0) \end{cases}}$$

8.2 RLC电路的自然响应（ Natural response of RLC circuits ）

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \qquad s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \qquad \boxed{\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}}$$

ω_0 :resonant frequency, 谐振频率

α: neper frequency ， damping factor 奈培频率，

阻尼系数

8.2 RLC电路的自然响应（ Natural response of RLC circuits ）

过阻尼(overdamped)：$\alpha > \omega_0$ ：

$$y = A1e^{s1t} + A2e^{s2t}$$

临界阻尼(critically damped)：$\alpha = \omega_0$ ：

$$y = (C1 + rC2) * e^{-\alpha t}$$

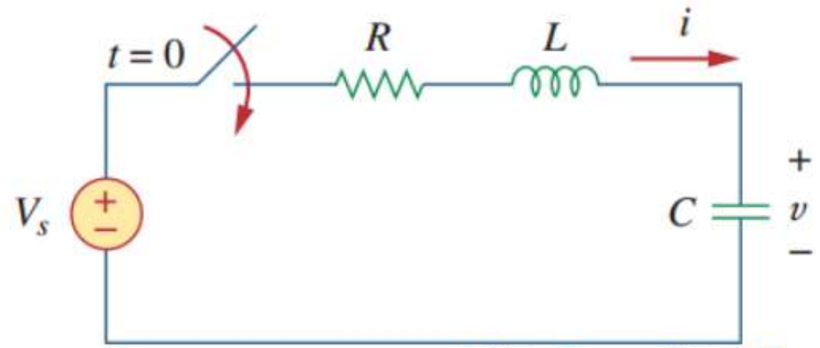
欠阻尼(underdamped) ：$\alpha < \omega_0$ ：

$$y = e^{-\alpha}\left(A1\cos(wdx) + A2\sin(wdx)\right) \quad wd = \sqrt{w0^2 - \alpha^2}$$

无阻尼振荡(undamped)：$\alpha = 0$

$$y = \left(A1\cos(w0x) + A2\sin(w0x)\right)$$

8.3 串联RLC电路阶跃响应



$i(0)=I_0$, $v(0)=V_0$

全响应=暂态响应+稳态响应

$$v(t) = v_t(t) + v_{ss}(t)$$

1.计算暂态响应：关闭电压源Vs，剩下的计算步骤同无源RLC电路

2.计算稳态响应：即当时间趋向无穷，v的最终值

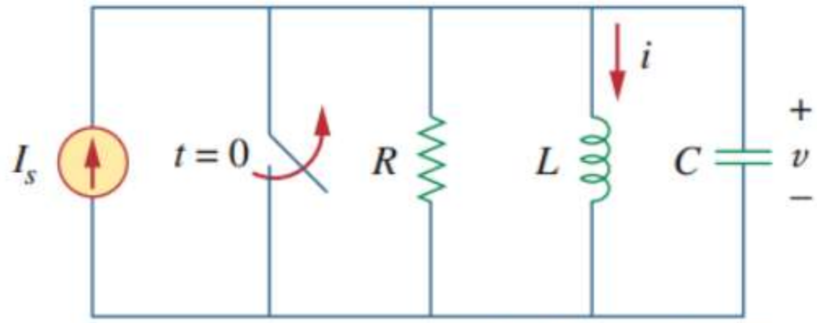
3.将暂态响应、稳态响应结果相加为最后结果

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(Overdamped)}$$

$$v(t) = V_s + (A_1 + A_2 t)e^{-\alpha t} \quad \text{(Critically damped)}$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t} \quad \text{(Underdamped)}$$

8.4 并联RLC电路阶跃响应



$i(0)=I_0$, $v(0)=V_0$

全响应=暂态响应+稳态响应

$$i(t) = i_t(t) + i_{ss}(t)$$

1.计算暂态响应：关闭电流源Is，剩下的计算步骤同无源RLC电路

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$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \qquad \text{(Overdamped)}$$

$$i(t) = I_s + (A_1 + A_2 t)e^{-\alpha t} \qquad \text{(Critically damped)}$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t} \qquad \text{(Underdamped)}$$

相量和功率分析

9-10

Chapter

9.1 正弦量和相量

9.2 phase shifters 移相器

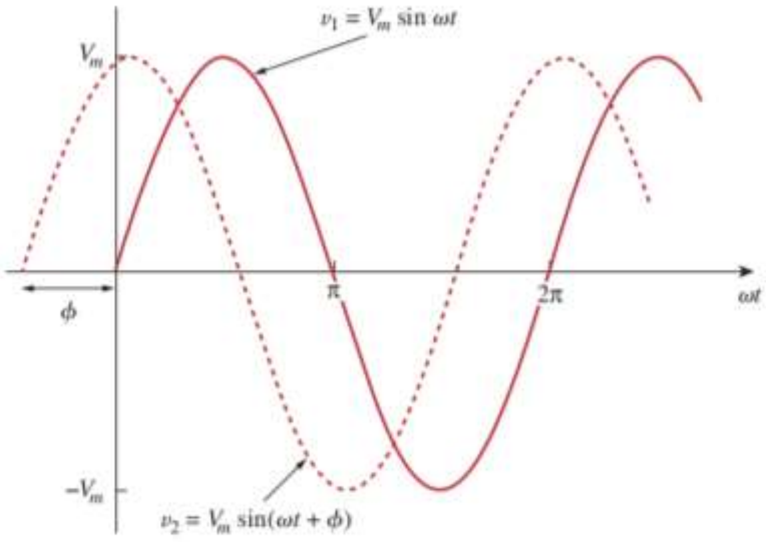
10.1 交流下的电路定律

10.2交流电路的功率表示

10.3 最大平均功率传输定理

9.1 正弦量(Sinusoids)

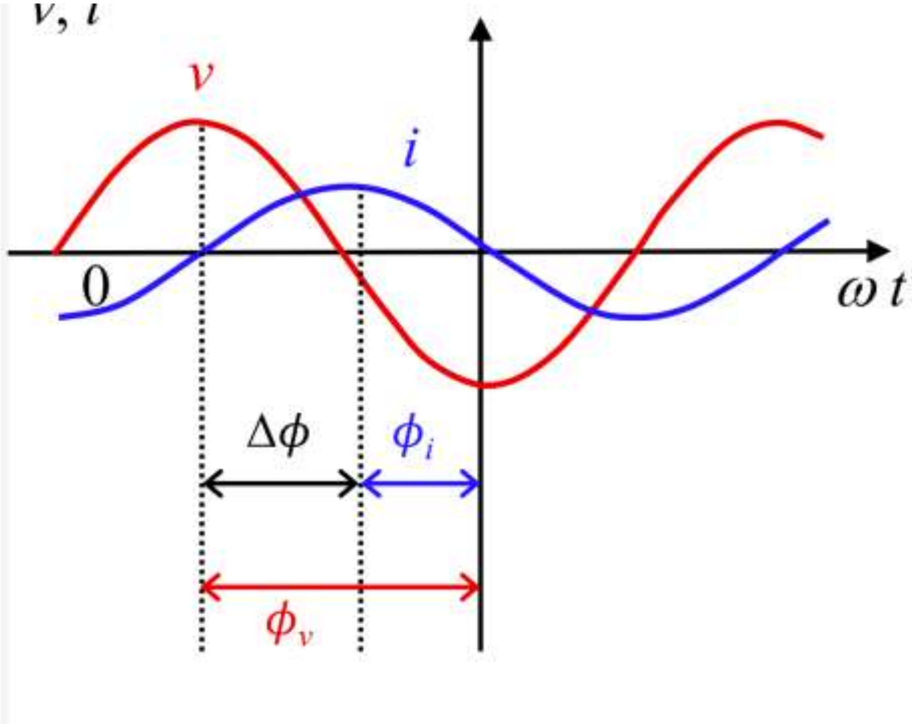
$$v(t) = V_m \sin(\omega t + \phi)$$



Vm: amplitude幅度
w: angular frequency角频率 rad/s
wt: argument 幅角

$T=\frac{2\pi}{w}$ $\quad f = \frac{1}{T}$ $\quad w=2\pi f$

9.1.2相位差(Phase Difference)



$$v = V_m \cos(\omega t + \phi_v) \text{ and } i = I_m \cos(\omega t + \phi_i)$$

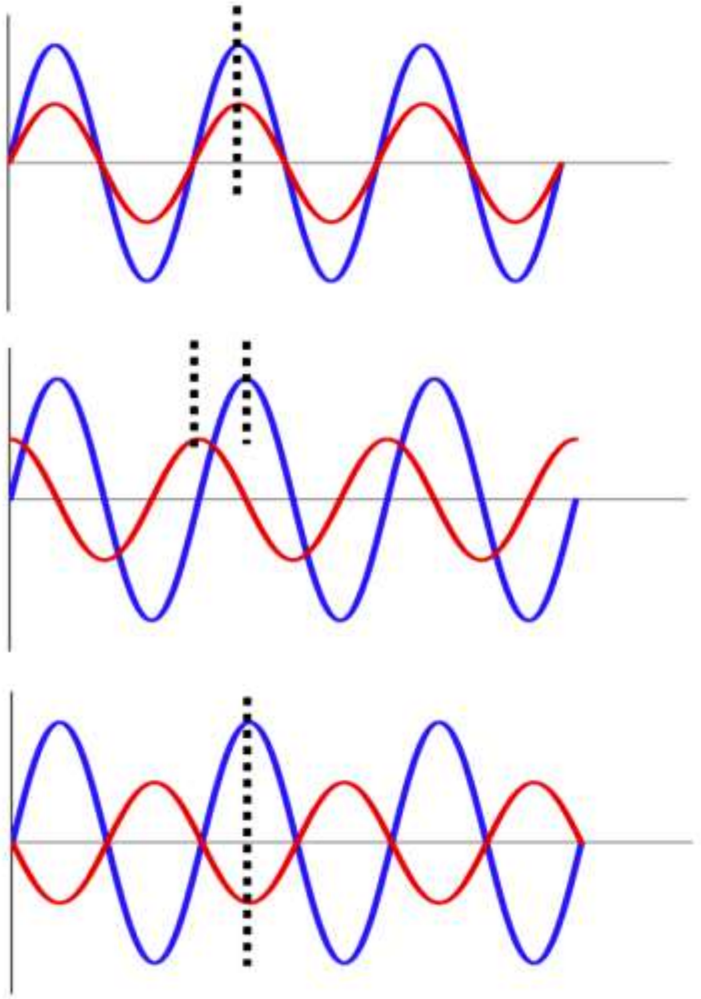
$$|\phi| \leq \pi$$

$$\Delta\phi = (\omega t + \phi_v) - (\omega t + \phi_i) = \phi_v - \phi_i$$

If $\Delta\phi = \phi_v - \phi_i > 0$, v leads i

If $\Delta\phi = \phi_v - \phi_i < 0$, i leads v

9.1.3相位差(Phase Difference)

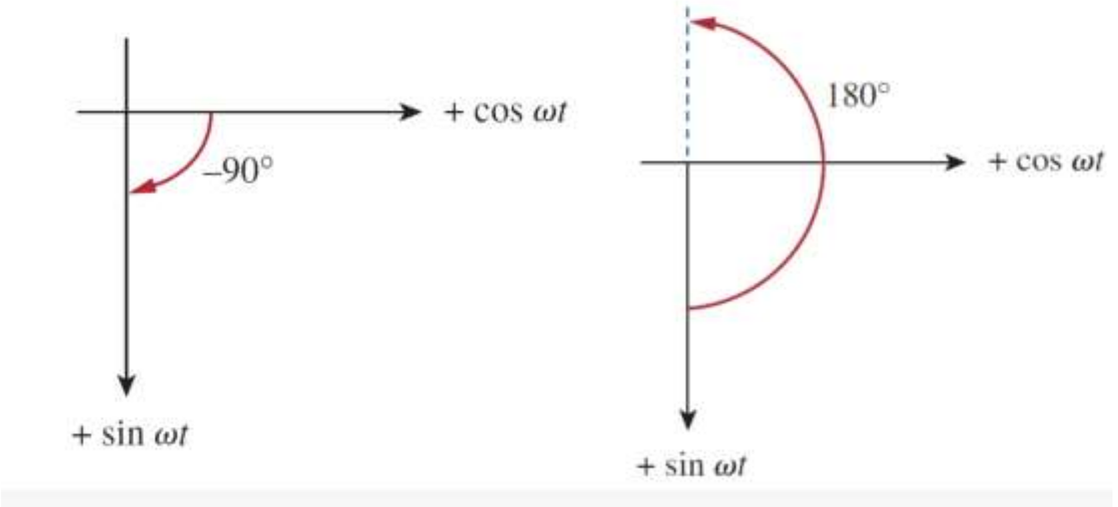


In phase
$\Delta\phi = 0$

Out of phase
$\Delta\phi \neq 0$

Antiphase
$\Delta\phi = \pi$

⬇ 逆时针为正角度



$$\frac{dv}{dt} \quad \Leftrightarrow \quad j\omega\mathbf{V}$$

(Time domain) **(Phasor domain)**

Does $jw\mathbf{V}$ lead or lag the $\mathbf{V}$?

Leading 90°

$$\int v\,dt \quad \Leftrightarrow \quad \frac{\mathbf{V}}{j\omega}$$

(Time domain) **(Phasor domain)**

Does $\mathbf{V}/jw$ lead or lag the $\mathbf{V}$?

Laging 90°

2 向量（Phasors)

$$z = x + jy$$

where $j = \sqrt{-1}$; x is the real part of z; y is the imaginary part of z.

$z = x + jy$ Rectangular form

$z = r\underline{/\phi}$ Polar form

$z = re^{j\phi}$ Exponential form

$\boxed{r = \sqrt{x^2 + y^2},}$ $\phi = \tan^{-1}\dfrac{y}{x}$

$$z = x + jy = r\underline{/\phi} = r(\cos\phi + j\sin\phi)$$

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 z_2 = r_1 r_2 \underline{/\phi_1 + \phi_2}$

Division: $\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}\underline{/\phi_1 - \phi_2}$

Reciprocal: $\dfrac{1}{z} = \dfrac{1}{r}\underline{/-\phi}$ $\Rightarrow$ $\dfrac{1}{j} = -j$

Square root: $\sqrt{z} = \sqrt{r}\,\underline{/\phi/2}$

Complex conjugate: $z^* = x - jy = r\underline{/-\phi} = re^{-j\phi}$

相角一般取-180º-180º

熟练掌握计算器复数计算功能

https://www.bilibili.com/video/av243860805/
https://www.bilibili.com/video/BV1xT4y1f7N7/

2 向量（Phasors)

向量不是实数，没有正负之分

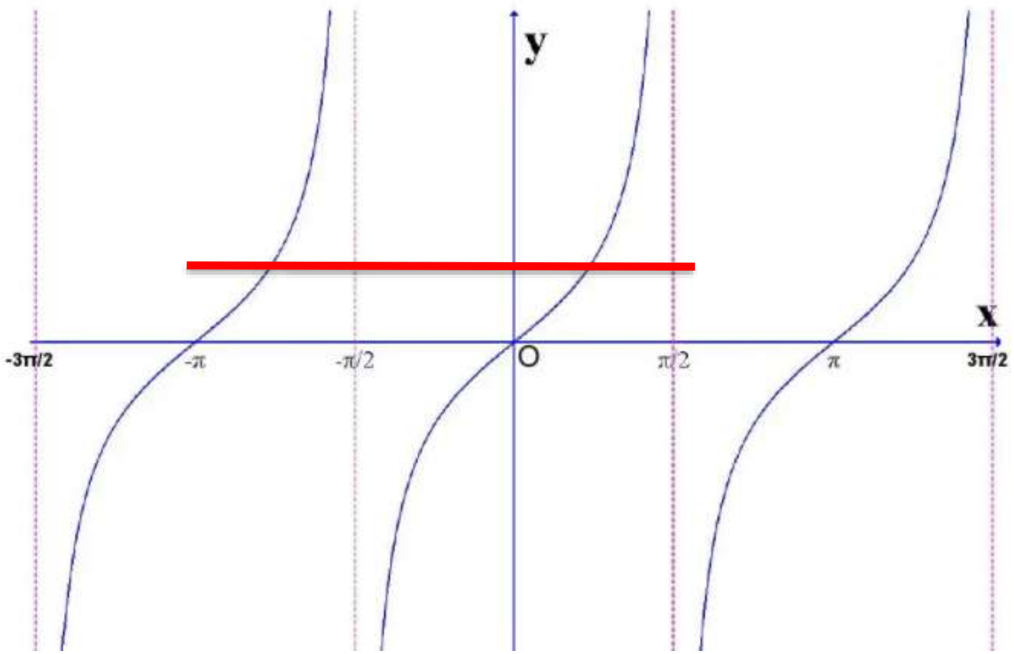
向量前加负号表示的是向量大小不变，方向取反，不是大小取负值（但是可以按照取负值的想法来进行运算)

$$30\angle 30° = 15\sqrt{3} + 15j = 30(\frac{\sqrt{3}}{2} + \frac{j}{2})$$

$$-15\sqrt{3} - 15j \neq 30\angle 30°$$

$$= 30(-\frac{\sqrt{3}}{2} - \frac{j}{2}) = 30\angle -150°$$

Y = tan x



2 向量（Phasors)

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi \Rightarrow \begin{aligned}\cos\phi &= \mathrm{Re}(e^{j\phi})\\ \sin\phi &= \mathrm{Im}(e^{j\phi})\end{aligned}$$

$$v(t) = V_m\cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m\underline{/\phi}$$

(Time-domain representation) (Phasor-domain representation)

$$v(t) = V_m\cos(\omega t + \phi) = \mathrm{Re}(V_m e^{j(\omega t+\phi)})$$

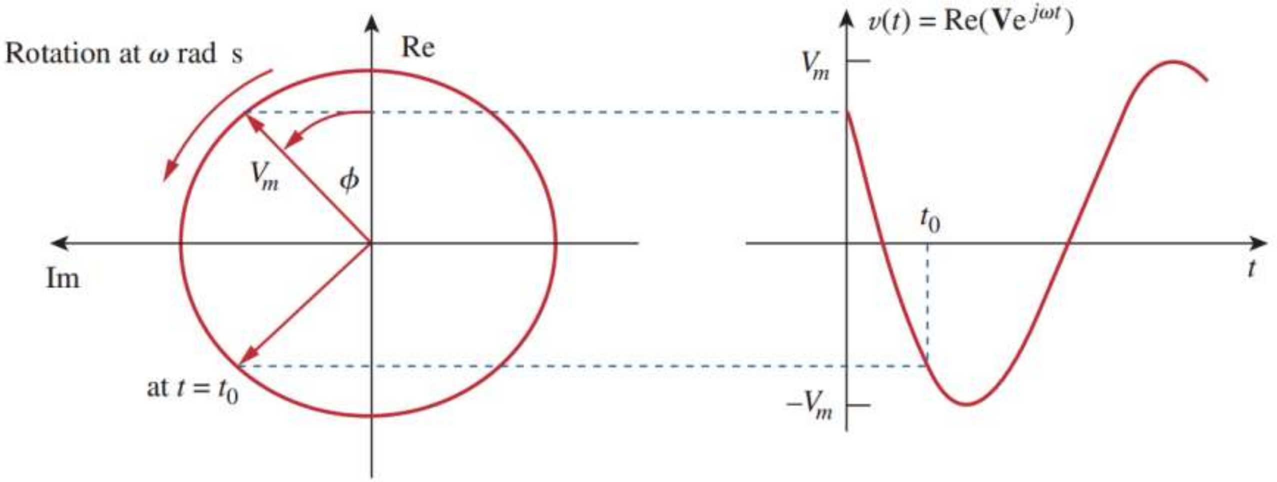
$$v(t) = \mathrm{Re}(V_m e^{j\phi} e^{j\omega t})$$

Thus,

$$\boxed{v(t) = \mathrm{Re}(\mathbf{V}e^{j\omega t})}$$

where

$$\mathbf{V} = V_m e^{j\phi} = V_m\underline{/\phi}$$



时域在转化为频域的时候隐藏掉了w的信息，因为w包含在向量旋转的角频率中了

2 向量（Phasors)

向量分析仅适用于频率恒定的情况，即只有当两个或多个正弦信号具有相同频率时，才能应用向量进行运算

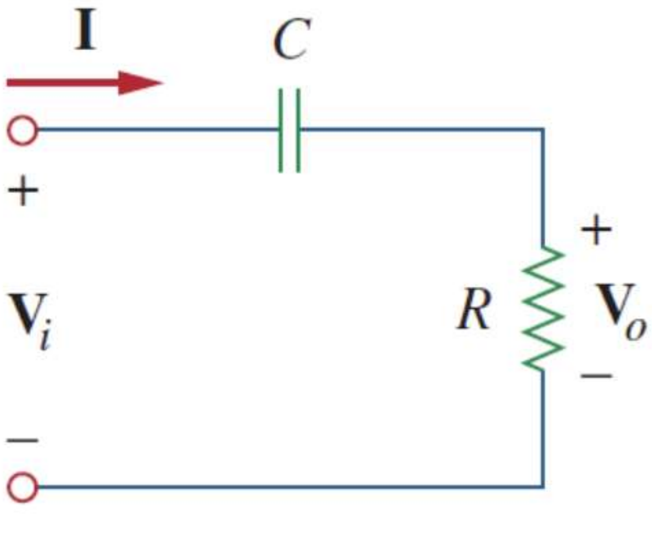
Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90°$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90°$

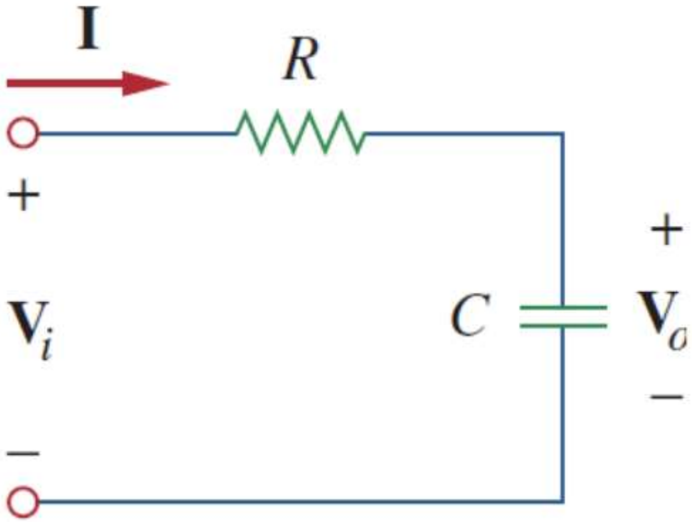
$$aj = a\angle 90^o$$

$$a = a\angle 0^o$$

9.2 移相器(Phase shifters) 不要求记忆，但是要会推导



(a)



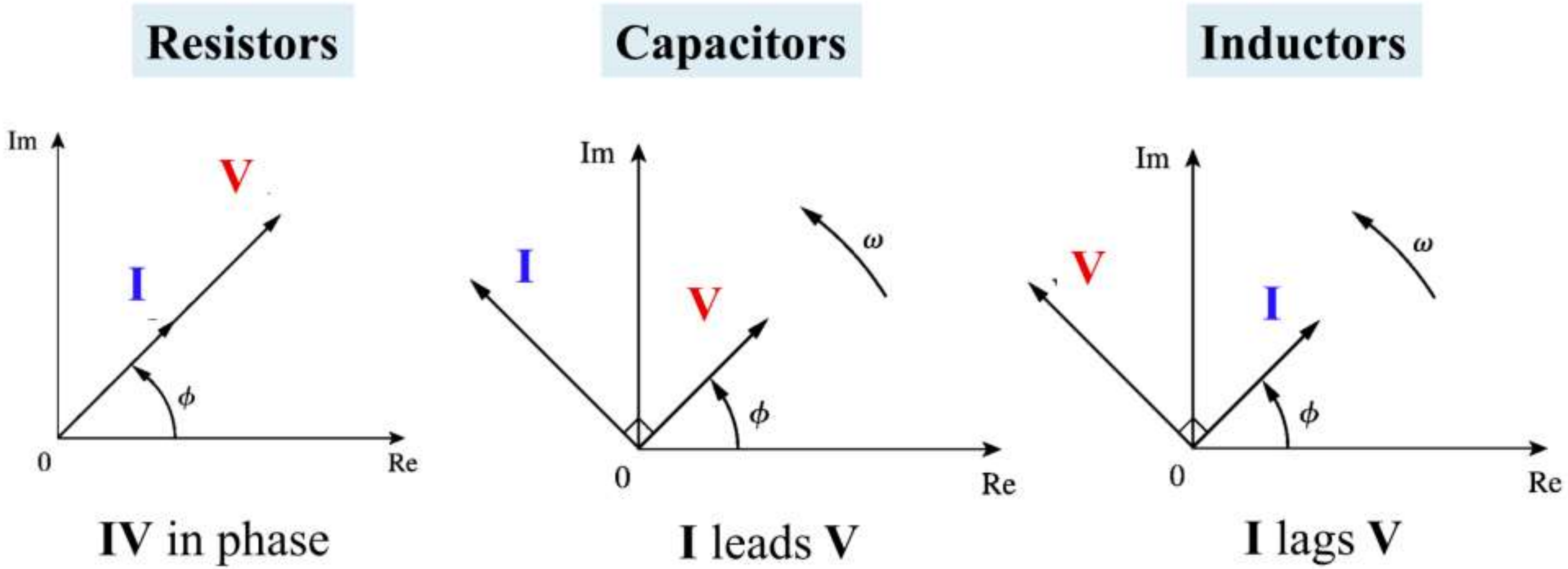
$$\theta = \tan^{-1}\left(\frac{1}{\omega RC}\right) = \tan^{-1}\left(\frac{-X_C}{R}\right)$$

$$\theta = \tan^{-1}(-\omega RC) = \frac{\pi}{2} - \tan^{-1}\left(\frac{X_C}{R}\right)$$

Positive phase shift

Negative phase shift

CIVIL



Resistors
Im
V
I
ϕ
0
Re
IV in phase
Capacitors
Im
ω
I
V
ϕ
0
Re
I leads V
Inductors
Im
ω
V
I
ϕ
0
Re
I lags V

10.1 交流下的电路定律

1.Nodal and mesh analysis

2.Superposition theorem and source transformation

3.Thevein and Norton equivalent circuits

4.AC circuits with op amp

分析方法同直流电路

10.2.1 交流电路的功率表示

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i)$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$p(t) = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

瞬时功率： Average Power:

P = $\frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) +$
$\frac{1}{2}V_m I_m \cos(2wt + \theta_v + \theta_i)$

平均功率： Average Power:

P = $\frac{1}{2}V_m I_m \cos(\theta_v - \theta_i)$

⬆ 取决于振幅和相差

10.2.2 有效值/均方根值 **Effective Value:**

$$I_{rms}=\frac{I_m}{\sqrt{2}} \qquad V_{rms}=\frac{V_m}{\sqrt{2}}$$

$$P=\frac{1}{2}V_m I_m\cos(\theta_v-\theta_i)=\frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos(\theta_v-\theta_i)=\boxed{V_{\mathrm{rms}}I_{\mathrm{rms}}\cos(\theta_v-\theta_i)}$$

$$P=I_{rms}^2 R\cos(\theta_v-\theta_i)=\frac{v_{rms}^2}{R}\cos(\theta_v-\theta_i)$$

V_{rms}* I_{rms} 视在功率 cos(θ_v-θ_i) 功率因数

10.2.3 有效值/均方根值 Effective Value:

$$I_{rms} = \frac{I_m}{\sqrt{2}} \qquad V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) = \boxed{V_{\rm rms} I_{\rm rms} \cos(\theta_v - \theta_i)}$$

$$P = I_{rms}^2 R cos(\theta_v - \theta_i) = \frac{v_{rms}^2}{R} cos(\theta_v - \theta_i)$$

V_{rms}* I_{rms} 视在功率 cos(θ_v-θ_i) 功率因数

Instantaneous power
瞬时功率

$$p(t) = \boxed{\frac{1}{2}V_m I_m \cos(\theta_v - \theta_i)} + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Average power
平均功率

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Effective values
有效值

$$V_{rms} = \frac{V_m}{\sqrt{2}}, \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

Power factor | 功率因数

$$\text{pf} = P/S = \cos(\theta_v - \theta_i)$$

Apparent power | 视在功率

$$S = V_{\text{rms}} I_{\text{rms}}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$

Instantaneous power
瞬时功率

$$p(t) = \boxed{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)} + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Average power
平均功率

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Effective values
有效值

$$V_{rms} = \frac{V_m}{\sqrt{2}}, \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

Power factor | 功率因数

$$\text{pf} = P/S = \cos(\theta_v - \theta_i)$$

Apparent power | 视在功率

$$S = V_{\text{rms}} I_{\text{rms}}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$

10.2.3 有功功率和无功功率 Real power and relative power

Since $\mathbf{Z} = R + jX$, $\mathbf{S} = I_{rms}^2 R + jI_{rms}^2 X = P + jQ$

P is the average or the real power: $P = Re[\mathbf{S}] = I_{rms}^2 R = V_{\mathrm{rms}} I_{\mathrm{rms}} \cos(\theta_v - \theta_i)$

Q is the reactive power: $Q = Im[\mathbf{S}] = I_{rms}^2 X = V_{\mathrm{rms}} I_{\mathrm{rms}} \sin(\theta_v - \theta_i)$

单位：瓦特

单位：VAR 乏

无功功率不消耗系统能量，只是用来衡量能量在电源和负载之间的转换，无功功率越大，转换越慢。

10.2.3 复功率,有功功率和无功功率 Complex power, real power and relative power

a) In analogy to voltage and current, we define a new phasor for power – **complex power**

$$S = \frac{1}{2}\mathbf{V}\mathbf{I}^* = \mathbf{V}_{\mathrm{rms}}\mathbf{I}_{\mathrm{rms}}^*$$

Where, $\mathbf{V}_{\mathbf{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{rms}\angle\theta_v$, $\mathbf{I}_{\mathbf{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{rms}\angle\theta_i$ are **effective voltage and current phasors.**

单位：V*A

Since $\mathbf{Z} = R + jX$, $\quad \mathbf{S} = I_{rms}^2 R + jI_{rms}^2 X = P + jQ$

P is the average or the real power: $P = Re[\mathbf{S}] = I_{rms}^2 R = V_{\mathrm{rms}} I_{\mathrm{rms}} \cos(\theta_v - \theta_i)$

单位：瓦特

Q is the reactive power: $Q = Im[\mathbf{S}] = I_{rms}^2 X = V_{\mathrm{rms}} I_{\mathrm{rms}} \sin(\theta_v - \theta_i)$

单位：VAR 乏

无功功率不消耗系统能量，只是用来衡量能量在电源和负载之间的转换。

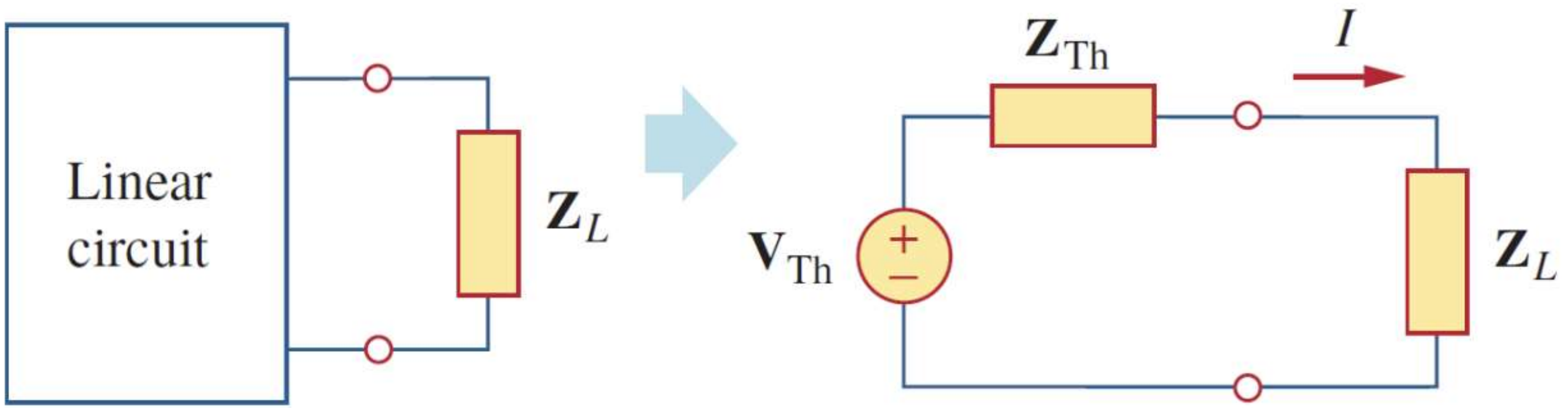
10. 2. 3 复功率, 有功功率和无功功率 Complex power, real power and relative power

$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$
$$= |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}| \underline{/\theta_v - \theta_i}$$
$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$
$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S\cos(\theta_v - \theta_i)$$
$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S\sin(\theta_v - \theta_i)$$
$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

10.3 最大平均功率传输定理 Maximum average power transfer

$$\mathbf{Z}_L = R_L + jX_L = R_{Th} - jX_{Th} = \mathbf{Z}_{Th}^*$$

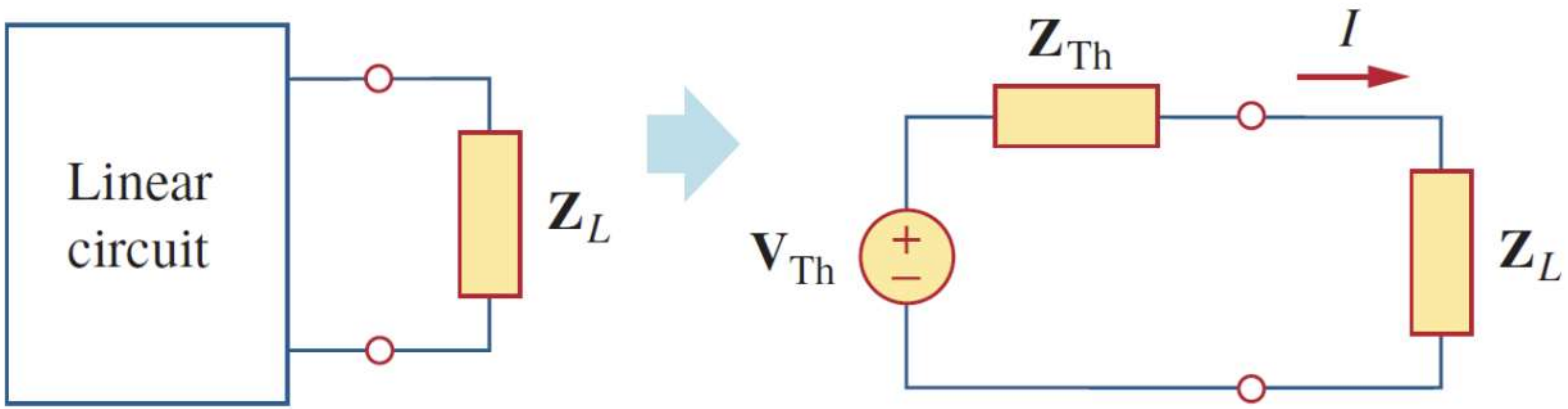
$$P_{max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}}$$



10.3 最大平均功率传输定理 Maximum average power transfer

$$\boxed{\mathbf{Z}_L = R_L + jX_L = R_{Th} - jX_{Th} = \mathbf{Z}_{Th}^*}$$

$$\boxed{P_{max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}}}$$

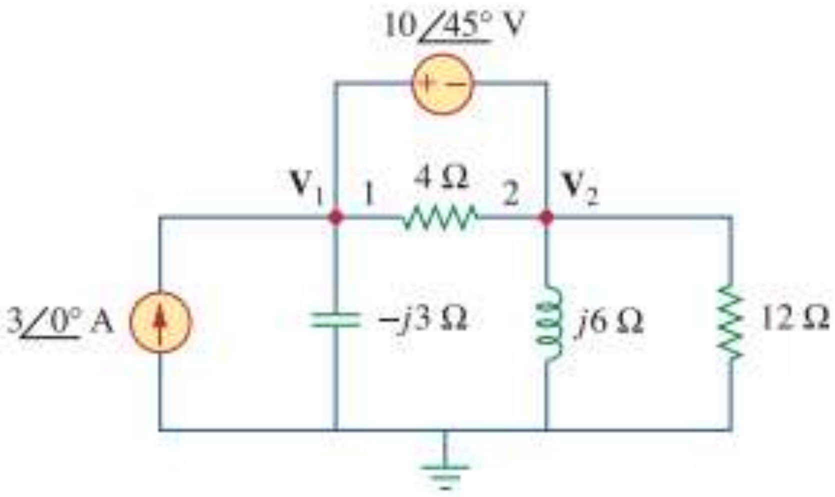


电路的频域分析注意事项：

注意陷阱：若题目给的频率单位为Hz，这个频率不是w(rad/s)，而是f，通过 w=2πf算出w

1.将电路转换到频域，写成向量形式（隐藏w）
2.利用电路分析法求解电路
3.将所求向量转换到时域，写出结果 (记得还原出w)
所有的电路定理仍然适用
题目所给量本来就是频域表达式时，结果也写为频域表达式，题目给时域，结果写成时域

Compute $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit of Fig. 10.4.



$$36 - 40\angle 135^\circ = (1 + j2)\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_2 = 31.41\angle -87.18^\circ \text{ V}$$

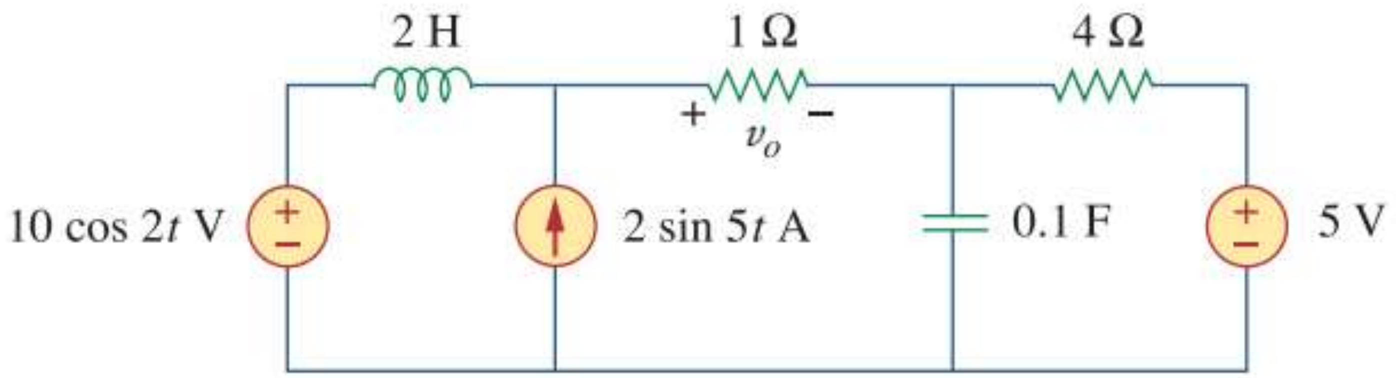
From Eq. (10.2.2),

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$

练习1

教材例10.6

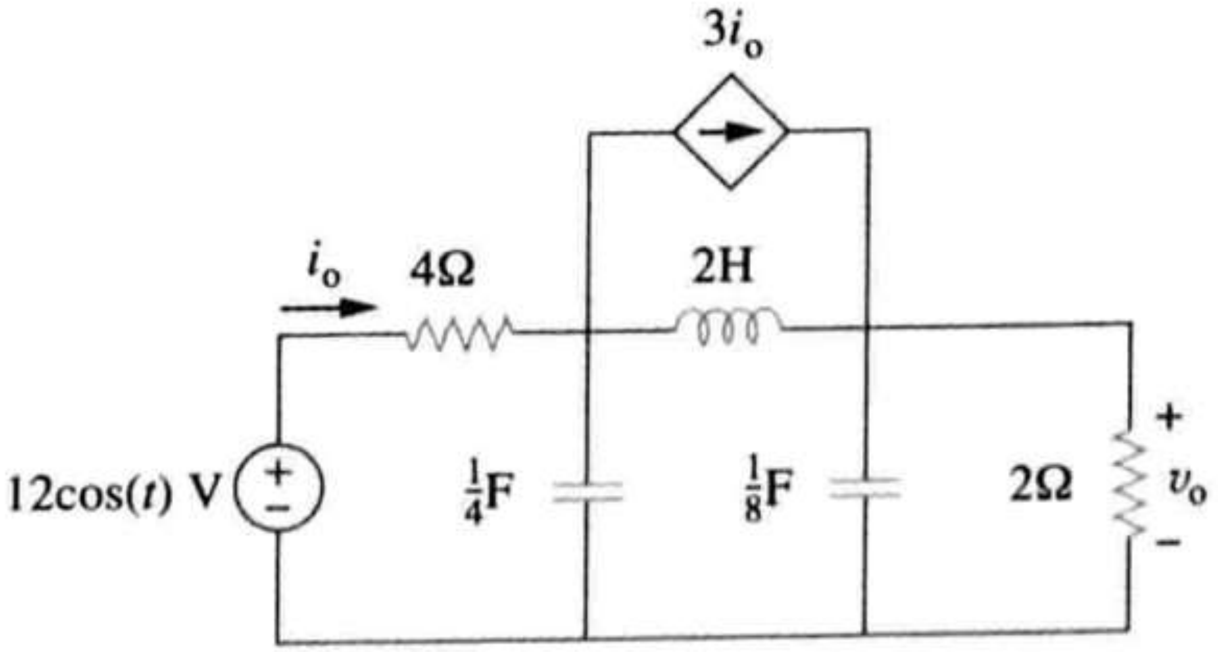
Find v_o of the circuit of Fig. 10.13 using the superposition theorem.



练习2

教材练习题10.62

62　利用戴维南定理计算图 10-105 所示电路中的 v_o。 **PS**



练习3

教材例11.9

例 11-9 当激励电压为 $v(t)=120\cos(100\pi t-20^\circ)$ V 时，流过某串接负载的电流为 $i(t)=4\cos(100\pi t+10^\circ)$ A，求该负载的视在功率与功率因数，并确定构成该串接负载的元件值。

练习4

教材例11.12

例 11-12 某负载从有效值 120V 的正弦电源中提取了 12kV·A 的功率，其功率因数为 0.856(滞后)，计算：(a)传递给该负载的平均功率与无功功率；(b)峰值电流；(c)负载阻抗。

频域分析

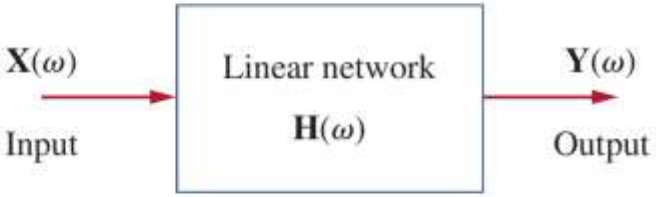
11

Chapter

频域相应 VS 时域响应

	时域响应	频域响应
分析的自变量	时间**t**	一般为角速度**w**
侧重点：	探究一个系统随时间的变化。	探究一个系统随频率变化的影响
本质上是两种处理方式，在地位上没有高低，但是时域很明显更加直观。		

11.1 基本概念



$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)}$$

Magnitude response
幅频响应

$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} = |H(\omega)|\, e^{j\varphi(x)}$$

phase response
相频响应

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1}{1 + j\omega RC}$$

幅频响应：$H = \dfrac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \dfrac{1}{\sqrt{1 + (\omega RC)^2}}$

相频响应：$\Delta\theta = -tan^{-1}(\omega RC)$

Input: voltage	Output: voltage	$\mathbf{H}(\omega)$ = Voltage gain = $\dfrac{\mathbf{V}_o(\omega)}{\mathbf{V}_i(\omega)}$	电压增益
Input: current	Output: current	$\mathbf{H}(\omega)$ = Current gain = $\dfrac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)}$	电流增益
Input: current	Output: voltage	$\mathbf{H}(\omega)$ = Transfer Impedance = $\dfrac{\mathbf{V}_o(\omega)}{\mathbf{I}_i(\omega)}$	转移阻抗
Input: voltage	Output: current	$\mathbf{H}(\omega)$ = Transfer Admittance = $\dfrac{\mathbf{I}_o(\omega)}{\mathbf{V}_i(\omega)}$	转移导纳

分贝表示法：

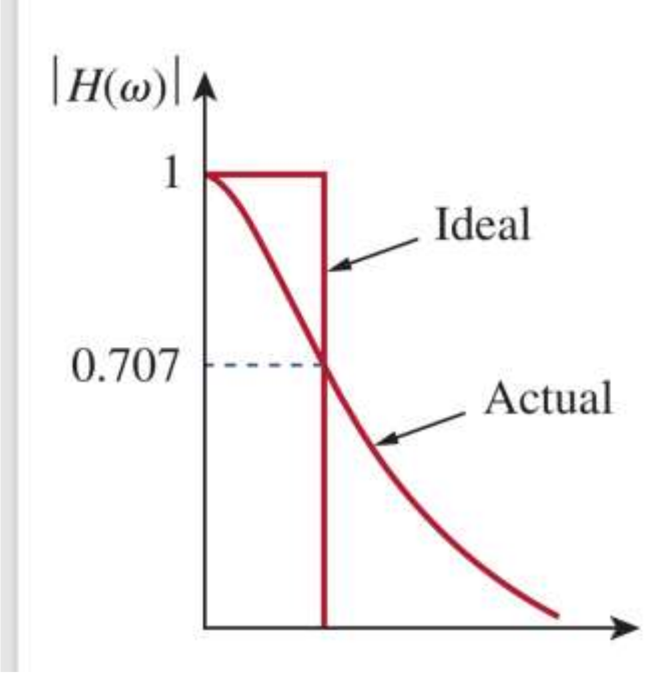
$$G_{\mathrm{dB}} = 10 \log_{10} \frac{P_2}{P_1}$$

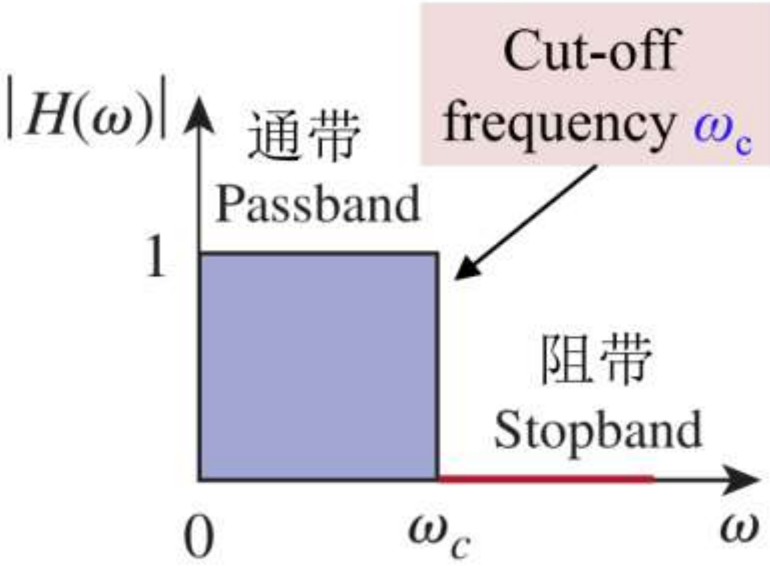
$$G_{\mathrm{dB}} = 20 \log_{10} \frac{V_2}{V_1} \qquad G_{\mathrm{dB}} = 20 \log_{10} \frac{I_2}{I_1}$$

11.2.1 低通滤波器 Low-pass filter

截止频率 (cut off frequencies): 使功率降为一半的频率点:

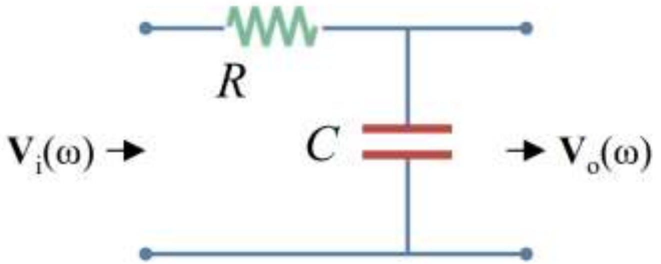
$$\text{即}|H(w)| = 1/\sqrt{2}\,|H(w)|\text{max}$$

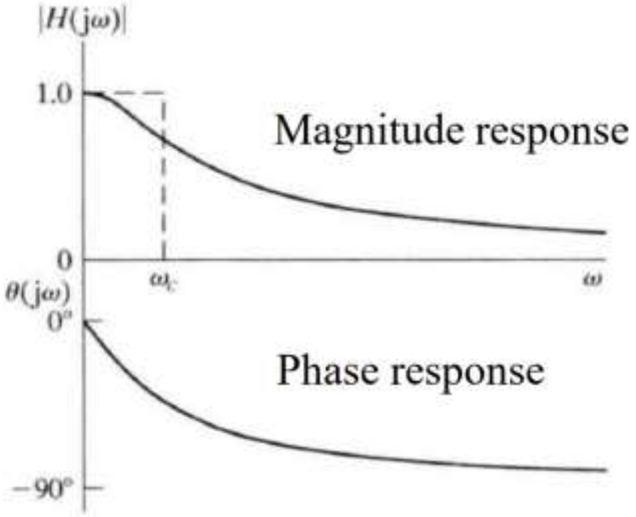




11.2.1 低通滤波器 Low-pass filter

- RC lowpass filters:





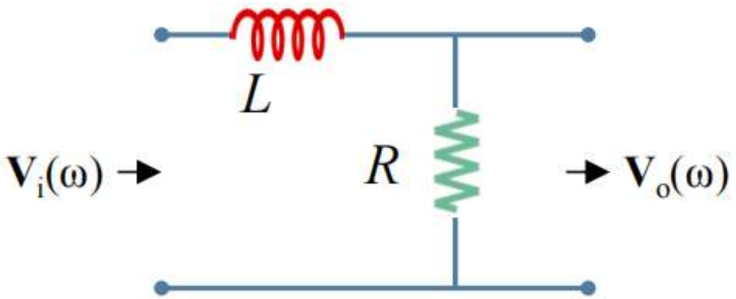
According to voltage division:

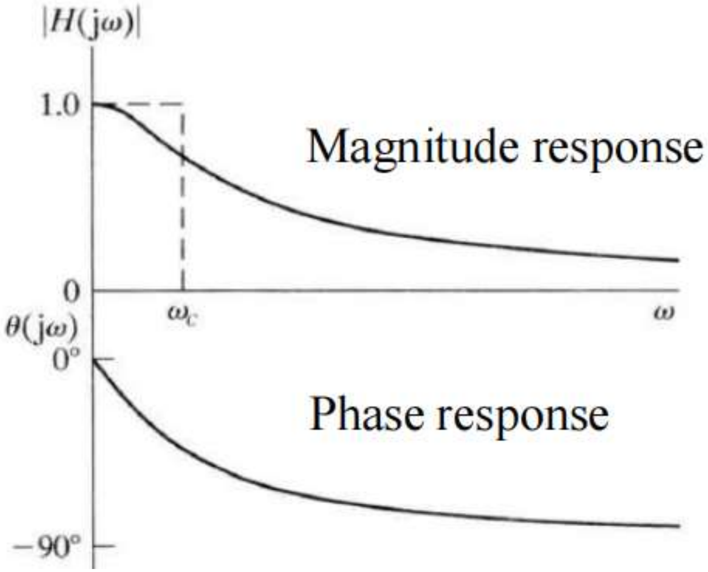
$$\mathbf{V_o} = \frac{1/j\omega C}{R + 1/j\omega C}\mathbf{V}_i \longrightarrow \begin{cases} \mathbf{H}(\omega) = \dfrac{1}{1 + j\omega RC} \\ |\mathbf{H}(\omega)| = \dfrac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \dfrac{1}{\sqrt{1 + (\omega RC)^2}} \\ \Delta\theta(\omega) = -tan^{-1}(\omega RC) \end{cases}$$

$$At\ \omega_c, |\mathbf{H}(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + 1}} = \frac{1}{\sqrt{2}} \longrightarrow \omega_c = \frac{1}{RC}$$

11.2.1 低通滤波器 Low-pass filter

- RL lowpass filters:



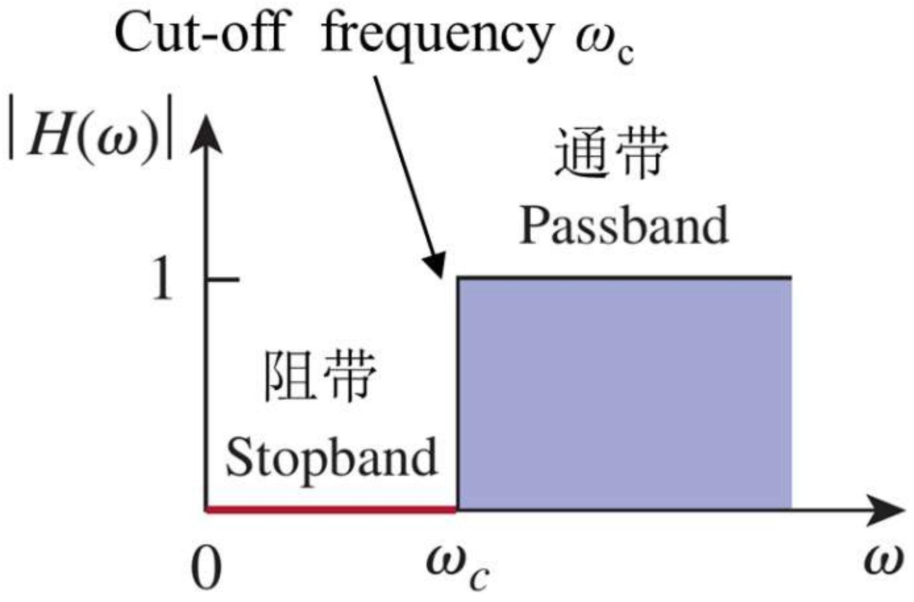


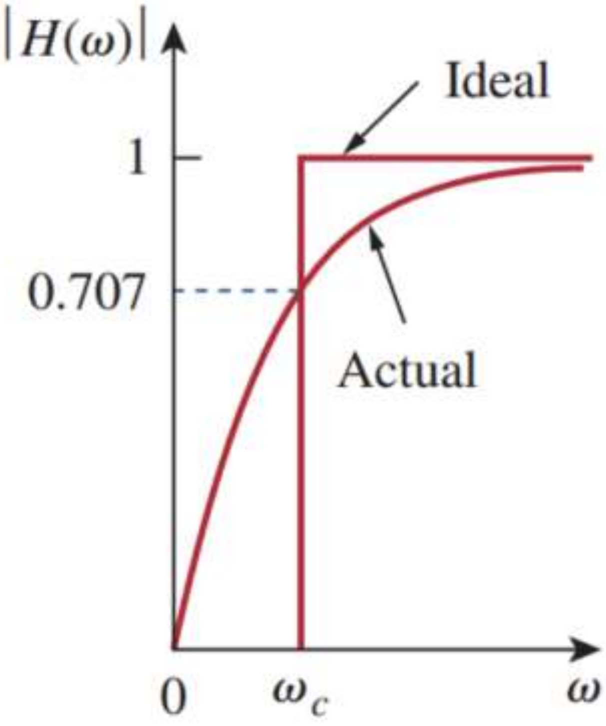
According to voltage division:

$$\mathbf{V_o} = \frac{R}{R + j\omega L}\mathbf{V}_i \longrightarrow \begin{cases} \mathbf{H}(\omega) = \dfrac{R/L}{R/L + j\omega} \\ |\mathbf{H}(\omega)| = \dfrac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \dfrac{R/L}{\sqrt{(R/L)^2 + \omega^2}} \\ \Delta\theta(\omega) = -tan^{-1}(\dfrac{\omega L}{R}) \end{cases}$$

$$At\ \omega_c, |\mathbf{H}(\omega)| = \frac{R/L}{\sqrt{(R/L)^2 + \omega_c^2}} = \frac{1}{\sqrt{2}} \longrightarrow \omega_c = \frac{R}{L}$$

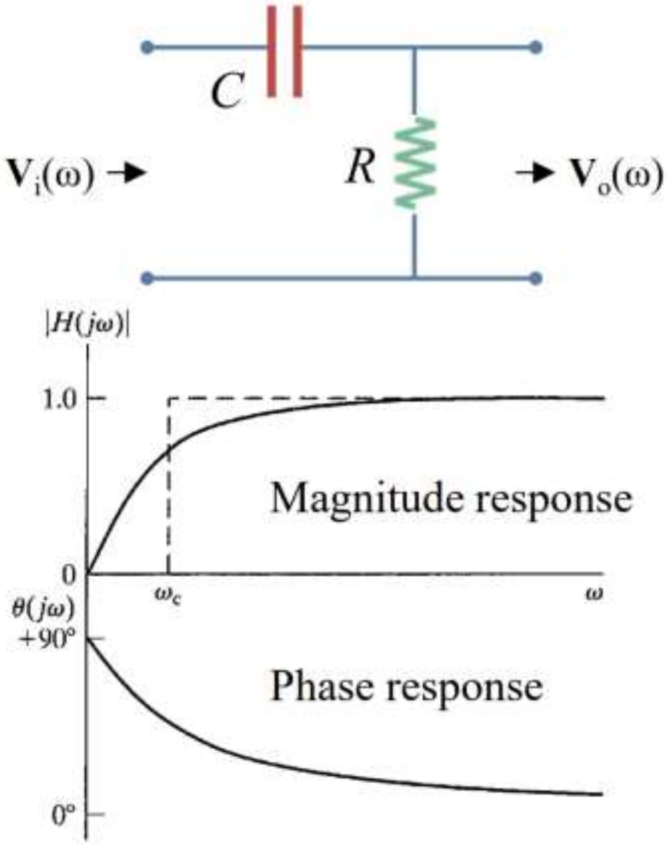
11.2.2 高通滤波器 High pass filter





11.2.2 高通滤波器　High pass filter

- RC highpass filters:



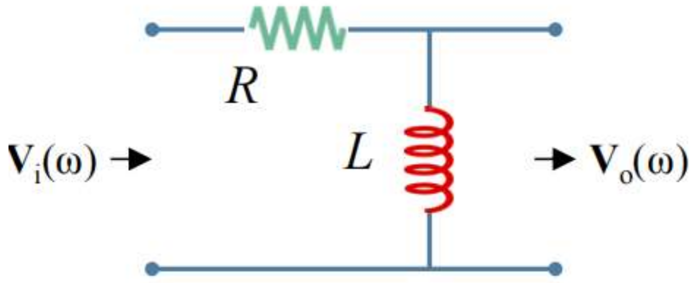
According to voltage division:

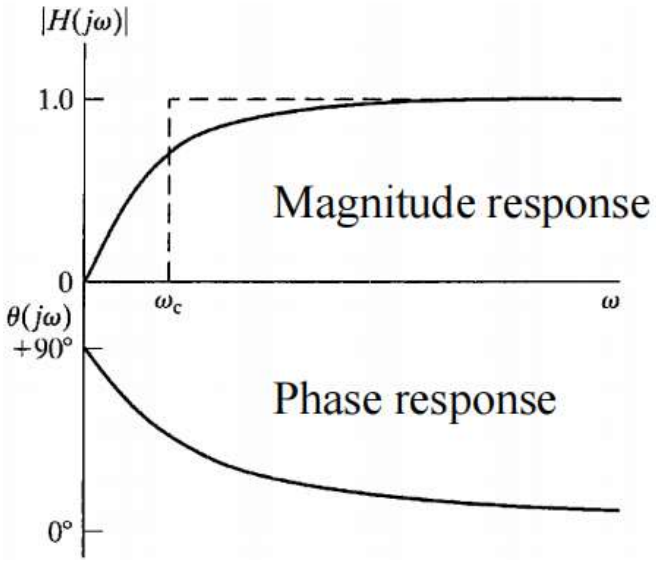
$$\mathbf{V_o} = \frac{R}{R + 1/j\omega C}\mathbf{V}_i \longrightarrow \begin{cases} \mathbf{H}(\omega) = \dfrac{j\omega RC}{1 + j\omega RC} \\ |\mathbf{H}(\omega)| = \dfrac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \dfrac{\omega RC}{\sqrt{1 + (\omega RC)^2}} \\ \Delta\theta(\omega) = tan^{-1}(1/\omega RC) \end{cases}$$

$$At\ \omega_c, |\mathbf{H}(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + 1}} = \frac{1}{\sqrt{2}} \longrightarrow \omega_c = \frac{1}{RC}$$

11.2.2 高通滤波器　High pass filter

- RL highpass filters:





According to voltage division:

$$\mathbf{V_o} = \frac{j\omega L}{R + j\omega L}\mathbf{V}_i \longrightarrow \begin{cases} \mathbf{H}(\omega) = \dfrac{j\omega}{R/L + j\omega} \\ |\mathbf{H}(\omega)| = \dfrac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \dfrac{\omega}{\sqrt{(R/L)^2 + \omega^2}} \\ \Delta\theta(\omega) = tan^{-1}(\dfrac{R}{\omega L}) \end{cases}$$

$$At\ \omega_c, |\mathbf{H}(\omega)| = \frac{R/L}{\sqrt{(R/L)^2 + \omega_c^2}} = \frac{1}{\sqrt{2}} \longrightarrow \omega_c = \frac{R}{L}$$

11.3　谐振频率，半功率频率点，带宽，品质因数

Δ谐振频率（w_o）：使所求量达到最大值所对应的频率。

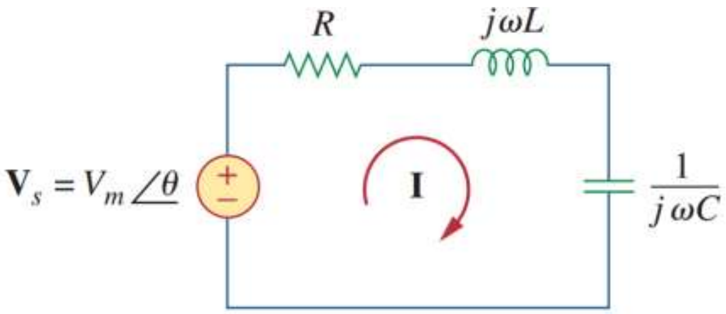
Δ半功率频率点（w_1, w_2）：使所求量为最大值$1/\sqrt{2}$所对应得频率。

Δ带宽（B）：两个半功率频率点的功率差，可以理解为可以通过滤波器的频域区间。

Δ品质因数（Q）：用于定义电路最大储存能量与一个周期电路消耗能量的比值，品质因数越大，表明这个电路损失的能量越小。

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

11.3.1 串联谐振



$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\mathrm{Im}(\mathbf{Z}) = \omega L - \frac{1}{\omega C} = 0$$

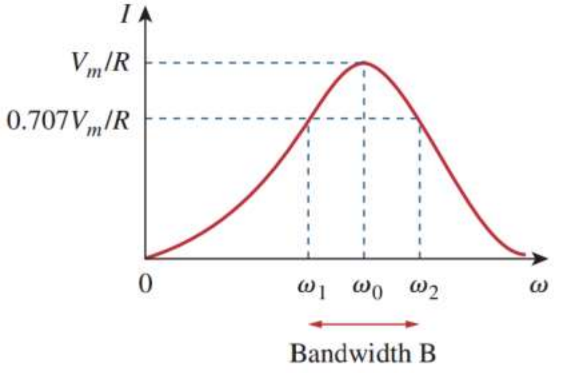
Resonant Frequency: $$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s} \qquad f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}}$$

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

ω_1 and ω_2 are called the *half-power frequencies*.

$$P(\omega_0) = \frac{1}{2}\frac{V_m^2}{R} \qquad P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$

$$\boxed{\begin{aligned} \omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \end{aligned}} \qquad \omega_0 = \sqrt{\omega_1\omega_2}$$

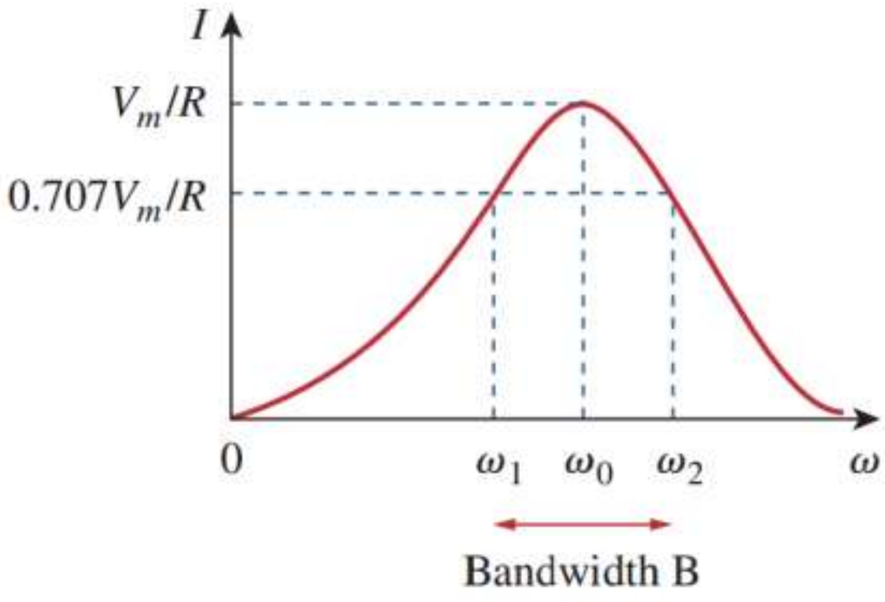


Band Width: $\boxed{B = \omega_2 - \omega_1}$

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

ω_1 and ω_2 are called the *half-power frequencies.*

$$P(\omega_0) = \frac{1}{2}\frac{V_m^2}{R} \qquad P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$



$$\boxed{\begin{aligned}\omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\end{aligned}}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

Band Width: $\boxed{B = \omega_2 - \omega_1}$

Quality factor $\boxed{Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}}$ $\boxed{B = \frac{R}{L} = \frac{\omega_0}{Q}}$

For high-Q circuit (Q>=10): $\boxed{\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2}}$

- Parallel Resonance 并联谐振

$$\mathbf{Y} = H(\omega) = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$\mathbf{Y} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

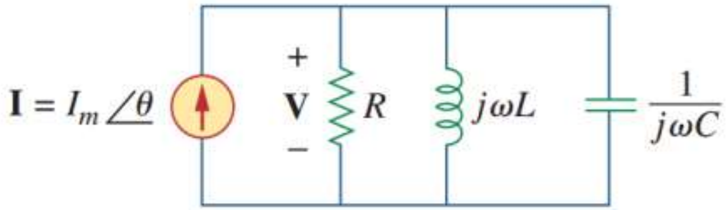
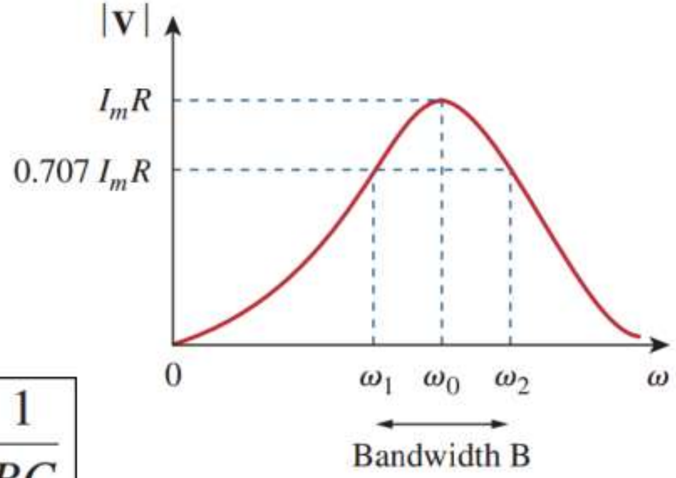


Figure 14.25
The parallel resonant circuit.

Resonance occurs when the imaginary part of **Y** is zero,

$$\omega C - \frac{1}{\omega L} = 0$$

Resonant Frequency: $\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}}$



$$\boxed{\begin{aligned}\omega_1 &= -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}\end{aligned}}$$

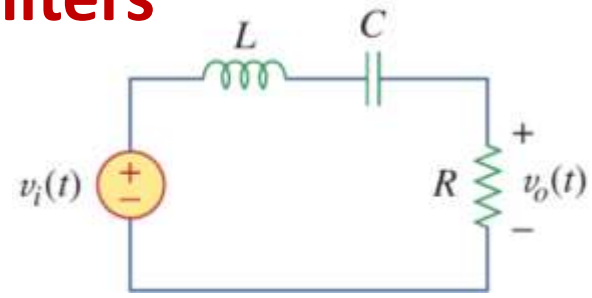
$$\boxed{B = \omega_2 - \omega_1 = \frac{1}{RC}}$$

$$\boxed{Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L}}$$

For high-Q circuit (Q>=10):

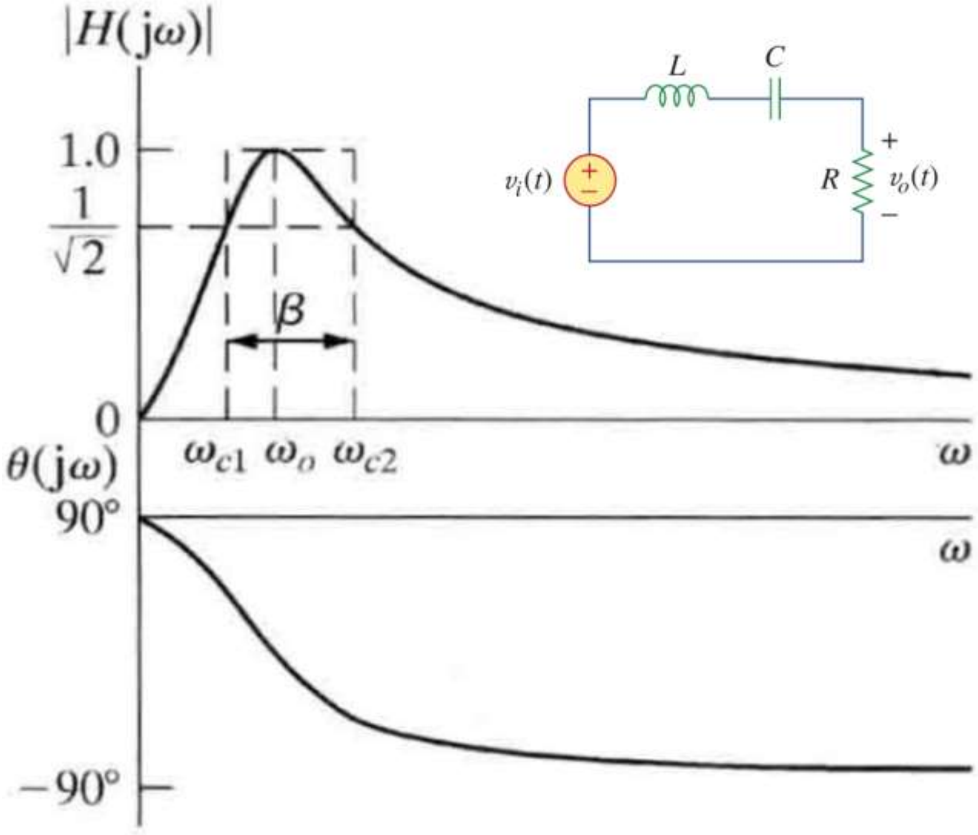
$$\boxed{\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2}}$$

11.4.1 带通滤波器 Bandpass filters



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



$$\omega_0 = \frac{1}{\sqrt{LC}}\,\text{rad/s} \qquad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

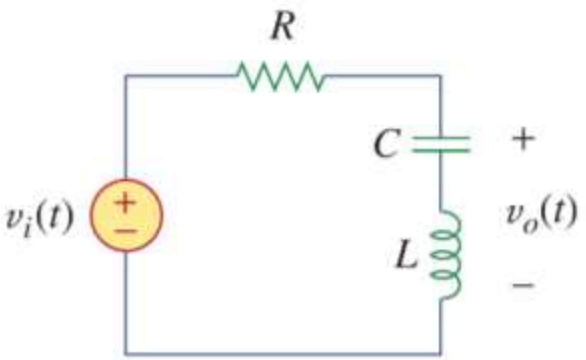
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_2 - \omega_1 \qquad B = \frac{R}{L} = \frac{\omega_0}{Q}$$

For high-Q circuit ($Q > 10$):

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

11.4.2 带阻滤波器 Band stop filters



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

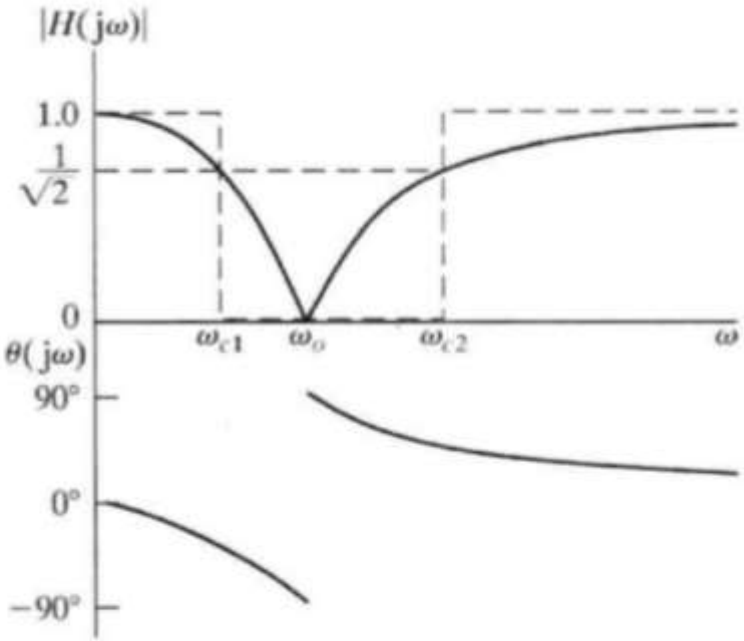
$$\omega_0 = \frac{1}{\sqrt{LC}}$$

a) ω_0 is also called frequency of rejection 抑制频率

b) $B = \omega_2 - \omega_1$ is called bandwidth of rejection 抑制带宽

$$|\mathrm{H}(\omega)| = \left|\frac{1}{LC} - \omega^2\right| \Big/ \sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \left(\frac{\omega R}{L}\right)^2}$$

$$\theta = -tan^{-1}\left(\frac{\omega R}{L} \Big/ \left(\frac{1}{LC} - \omega^2\right)\right)$$



$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s} \qquad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

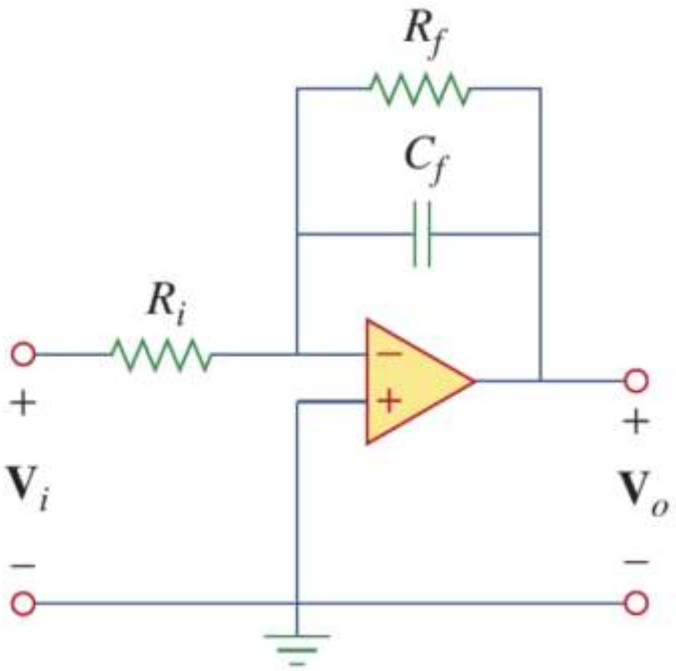
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

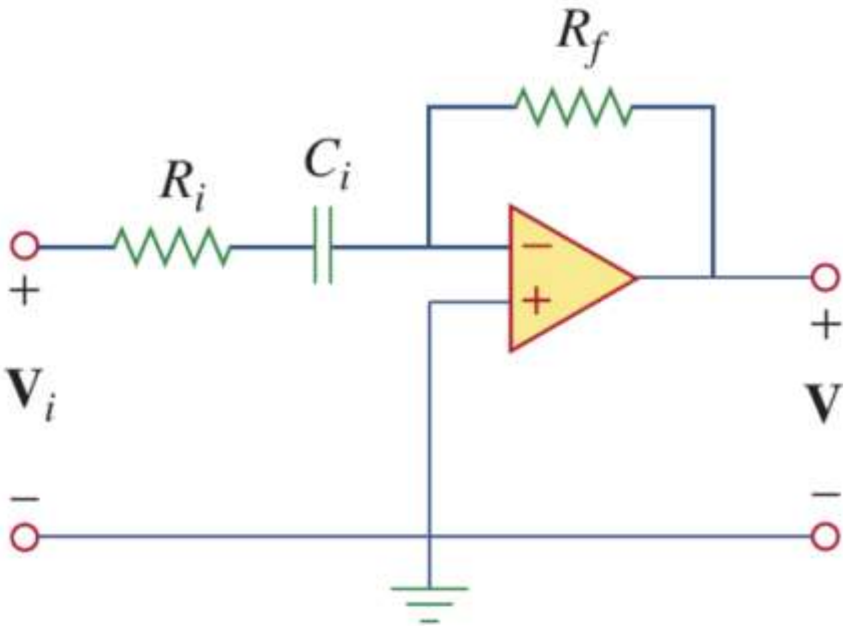
$$B = \omega_2 - \omega_1 \qquad B = \frac{R}{L} = \frac{\omega_0}{Q}$$

For high-Q circuit ($Q > 10$):

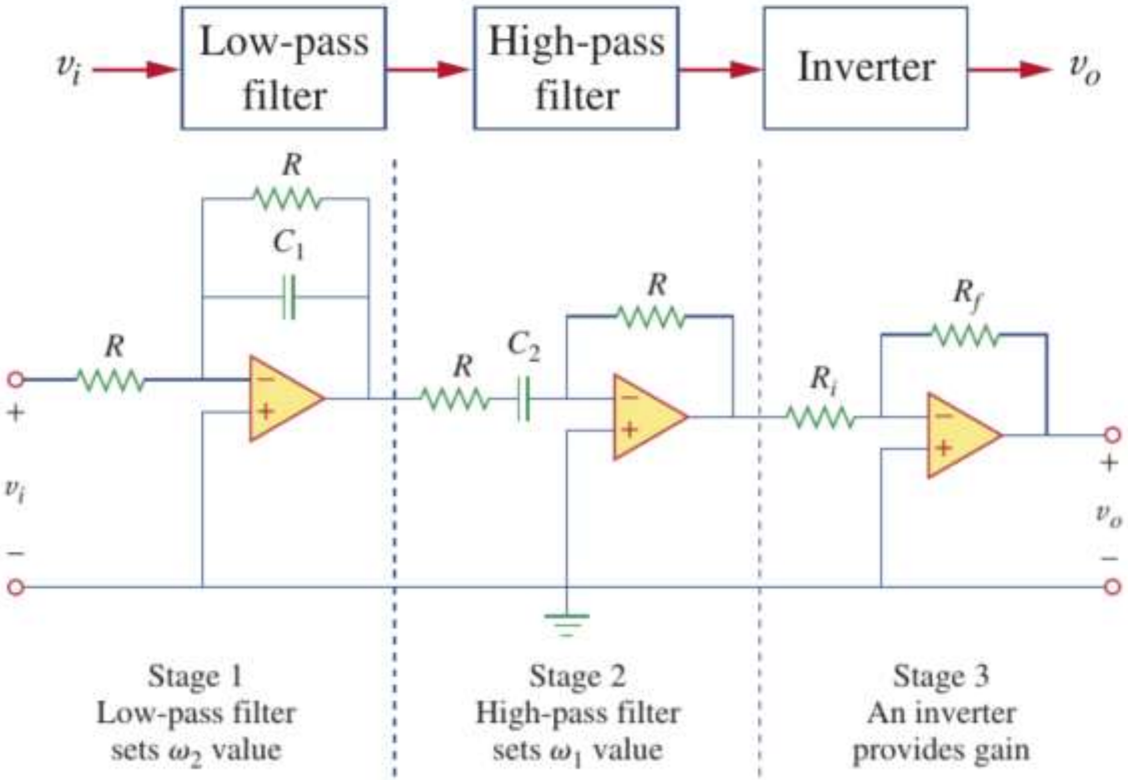
$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \; \omega_2 \simeq \omega_0 + \frac{B}{2}$$

11.5 有源滤波器　Active filters

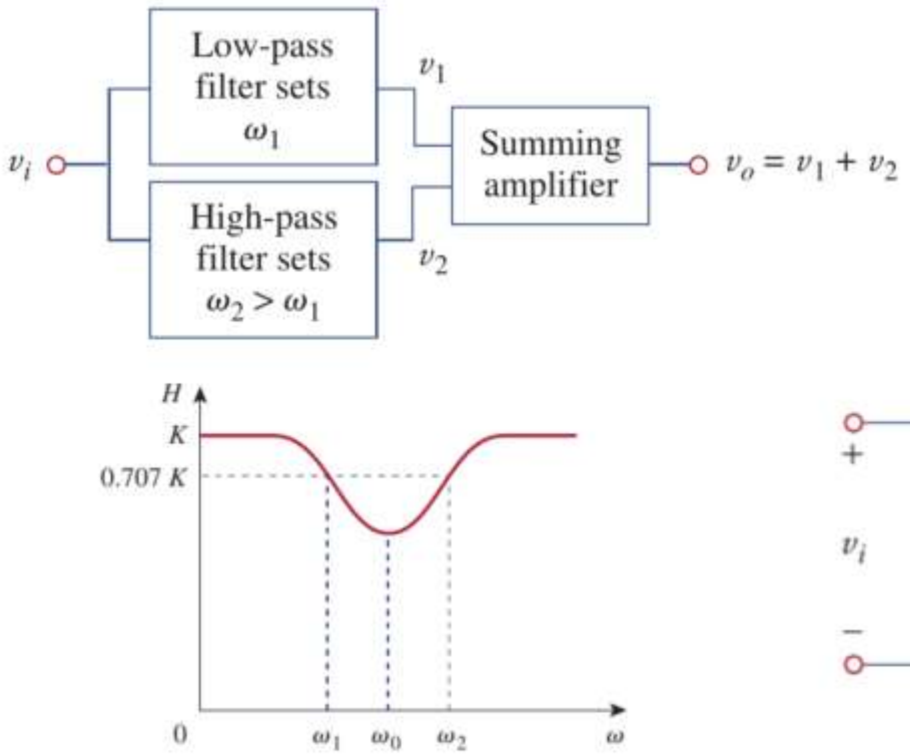


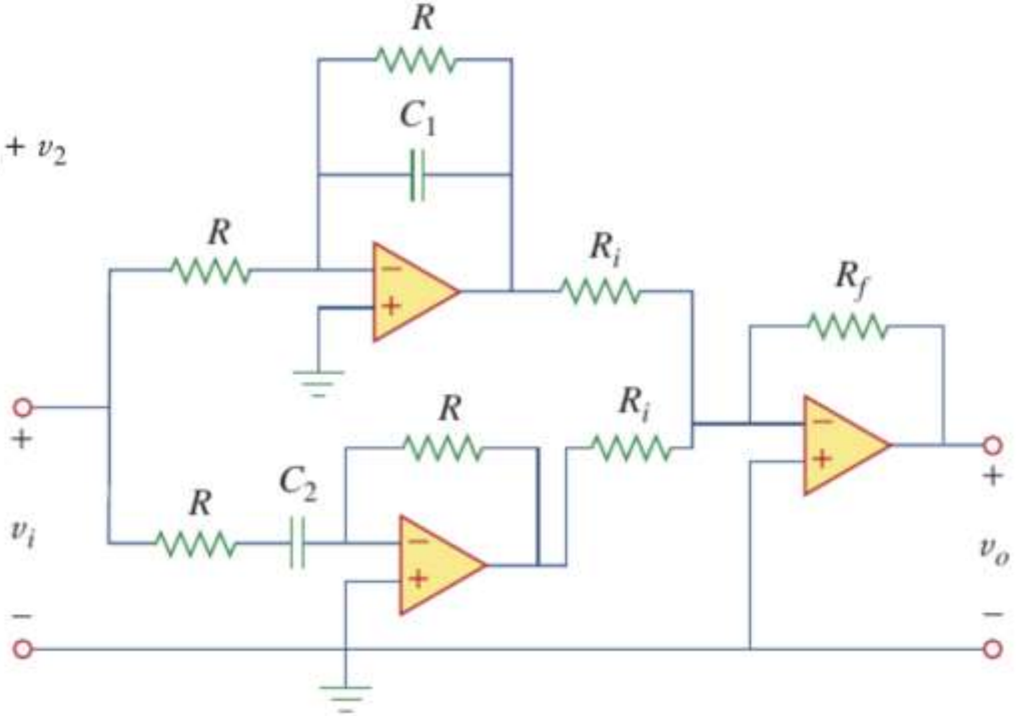


11.5 有源滤波器 Active filters



11.5 有源滤波器 Active filters





如何求解滤波器类型的题目？

前面所有的电路模型其实都不用背记！如果遇到复杂电路，常见处理方式：

Δ **Step1:** 判断滤波器类型、。假设 $\omega \to 0$ （此时电感可以看作导线，电容看作断路，判断输出值；再假设$\omega \to \infty$ （此时电感可以看作断路，电容可以看作导线）。

Δ **Step2:** 将题目所给量转换为频域量，直接求解H（w）。

Δ **Step3:** 写出其幅频响应和相频响应表达式。

Δ **Step4:** 利用定义求幅频响应最大值及其频率，即为wo;（本质求函数最大值）

Δ **Step5:** 令幅频响应值为最大值的$1/\sqrt{2}$，求解w1,w2，进而算出带宽。

Δ **Step6:**按照题目要求求解其他物理量。

Example 14.10

Determine what type of filter is shown in Fig. 14.39. Calculate the corner or cutoff frequency. Take $R = 2\ \text{k}\Omega$, $L = 2$ H, and $C = 2\ \mu\text{F}$.

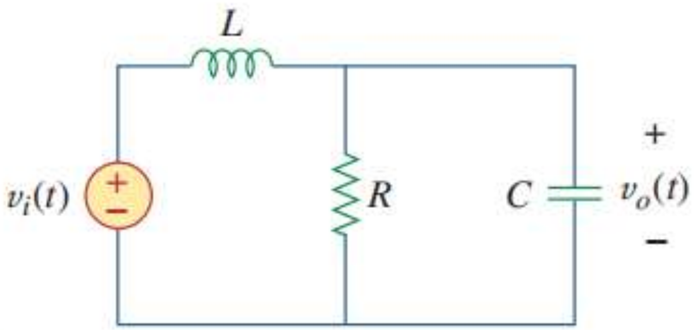


Figure 14.39
For Example 14.10.

14.57 Determine the center frequency and bandwidth of the bandpass filters in Fig. 14.88.

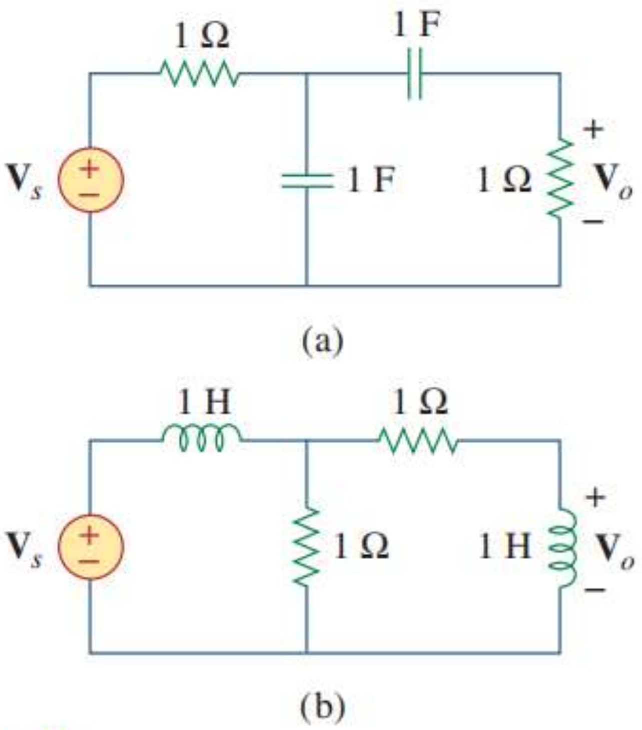


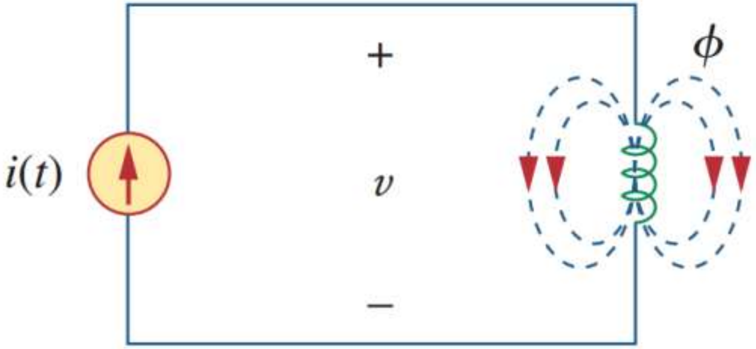
Figure 14.88
For Prob. 14.57.

磁耦合电路

12

Chapter

12.1 自感(Self Inductance)



N匝线圈，流过电流i，产生磁通量ϕ，则线圈中感应电压：

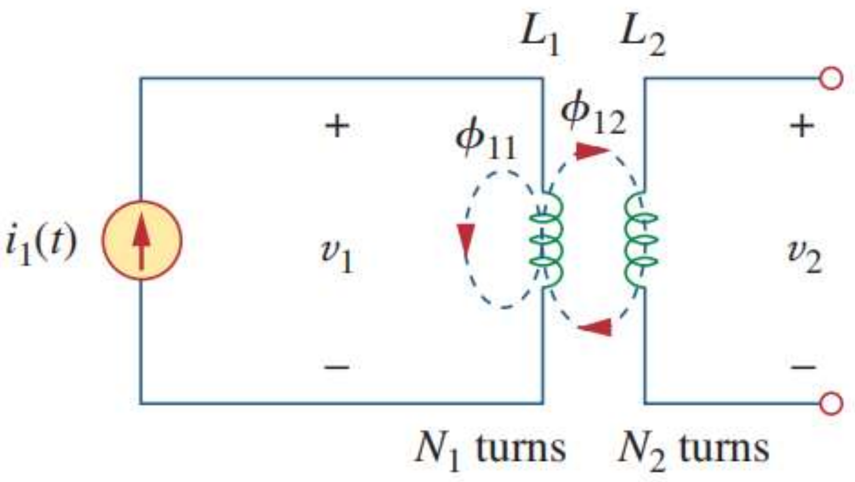
$$v = N\frac{d\phi}{dt}$$

$$v = N\frac{d\phi}{di}\frac{di}{dt}$$

$$v = L\frac{di}{dt} \qquad L = N\frac{d\phi}{di}$$

L: self-inductance自感

12.2 互感(Mutual Inductance)



线圈2本身没有电流，线圈1的磁通量由两部分构成：

$$\phi_1 = \phi_{11} + \phi_{12}$$

Φ_{11}仅与线圈1交链， Φ_{12}与两个线圈交链，

$$v_1 = N_1 \frac{d\phi_1}{dt}$$

$$v_1 = N_1 \frac{d\phi_1}{di_1}\frac{di_1}{dt} = L_1 \frac{di_1}{dt}$$

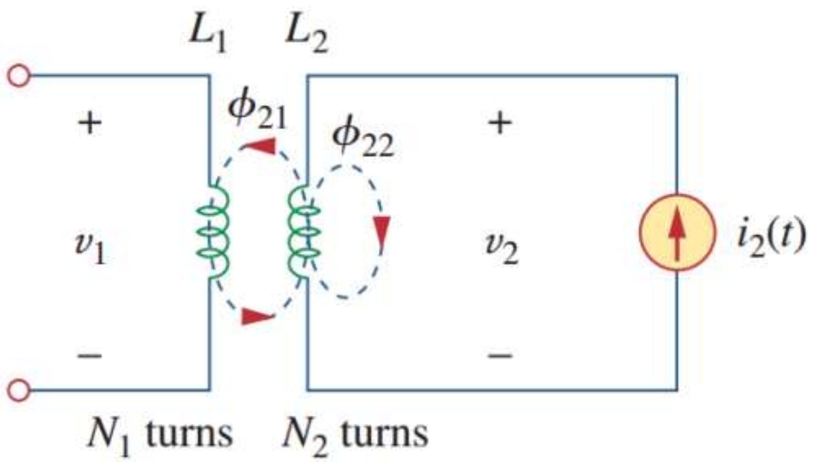
$$v_2 = N_2 \frac{d\phi_{12}}{dt}$$

$$v_2 = N_2 \frac{d\phi_{12}}{di_1}\frac{di_1}{dt} = M_{21}\frac{di_1}{dt}$$

$$\boxed{M_{21} = N_2 \frac{d\phi_{12}}{di_1}}$$

M_{21}：线圈2相对于线圈1的互感

12.2互感(Mutual Inductance)



线圈1本身没有电流，线圈2的磁通量由两部分构成：

$$\phi_2 = \phi_{21} + \phi_{22}$$

Φ_{22}仅与线圈2交链，Φ_{21}与两个线圈交链

$$v_2 = N_2\frac{d\phi_2}{dt} = N_2\frac{d\phi_2}{di_2}\frac{di_2}{dt} = L_2\frac{di_2}{dt}$$

$$v_1 = N_1\frac{d\phi_{21}}{dt} = N_1\frac{d\phi_{21}}{di_2}\frac{di_2}{dt} = M_{12}\frac{di_2}{dt}$$

$$M_{12} = N_1\frac{d\phi_{21}}{di_2}$$

M_{12}：线圈1相对于线圈2的互感

$$\boxed{M_{12} = M_{21} = M}$$

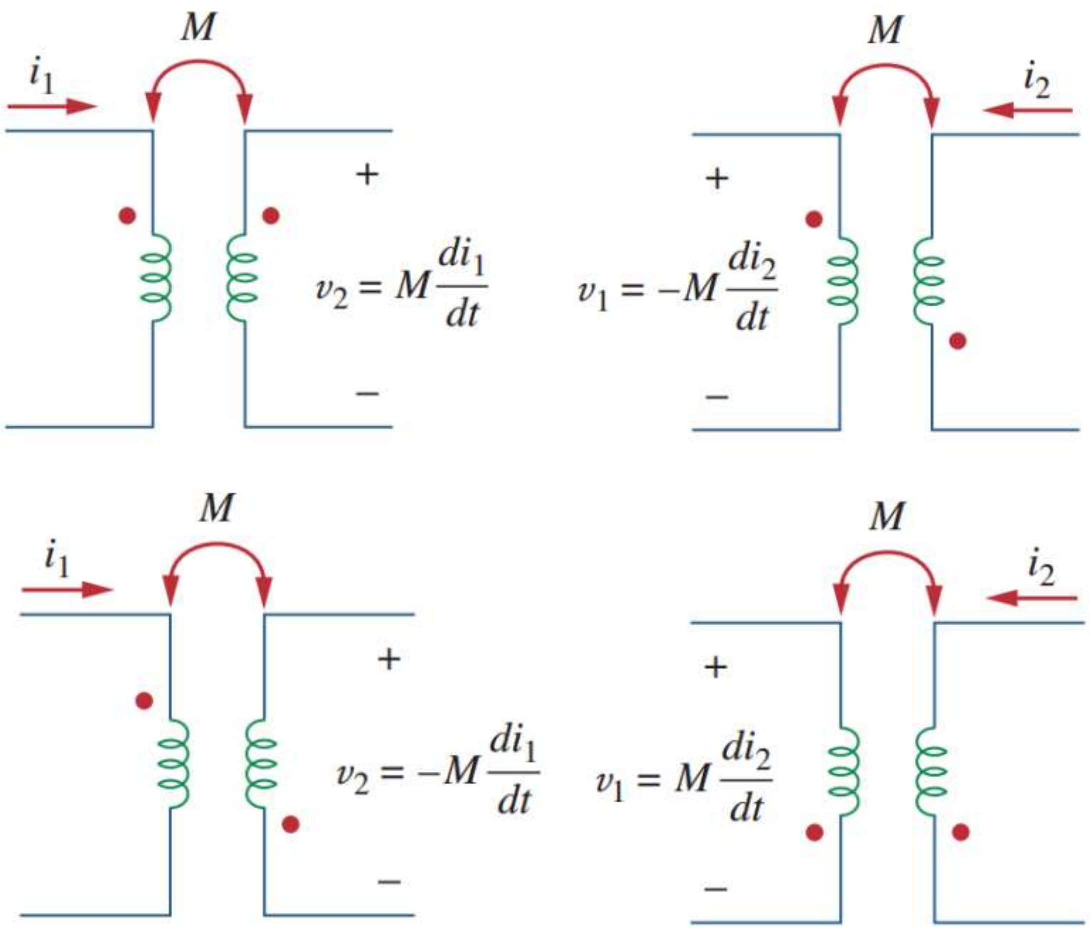
单位：H

12.2.2 同名端规则

同名端规则：

如果电流进入一个线圈的同名端，则在第二个线圈的同名端处，互感电压的参考极性为正。

如果电流从一个线圈的同名端流出，则在第二个线圈的同名端处，互感电压的参数极性为负。

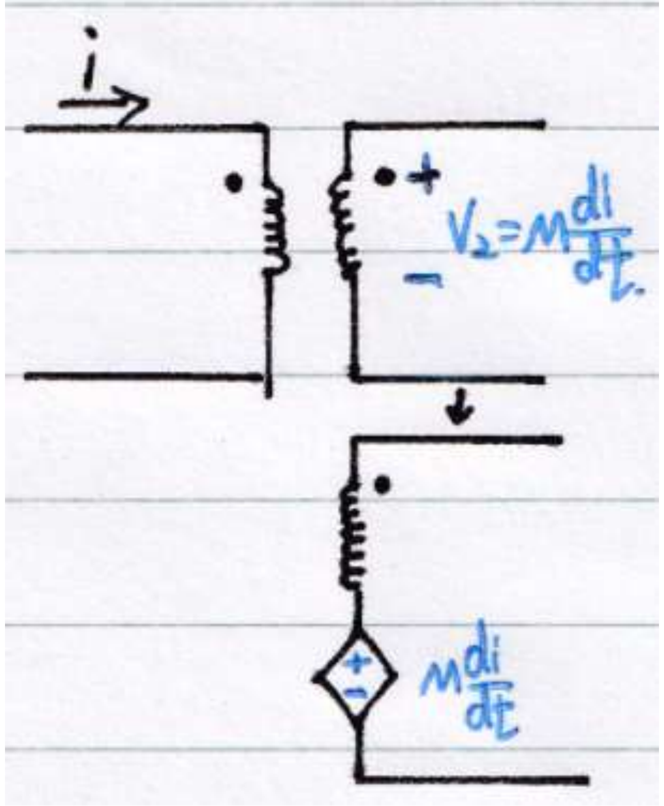


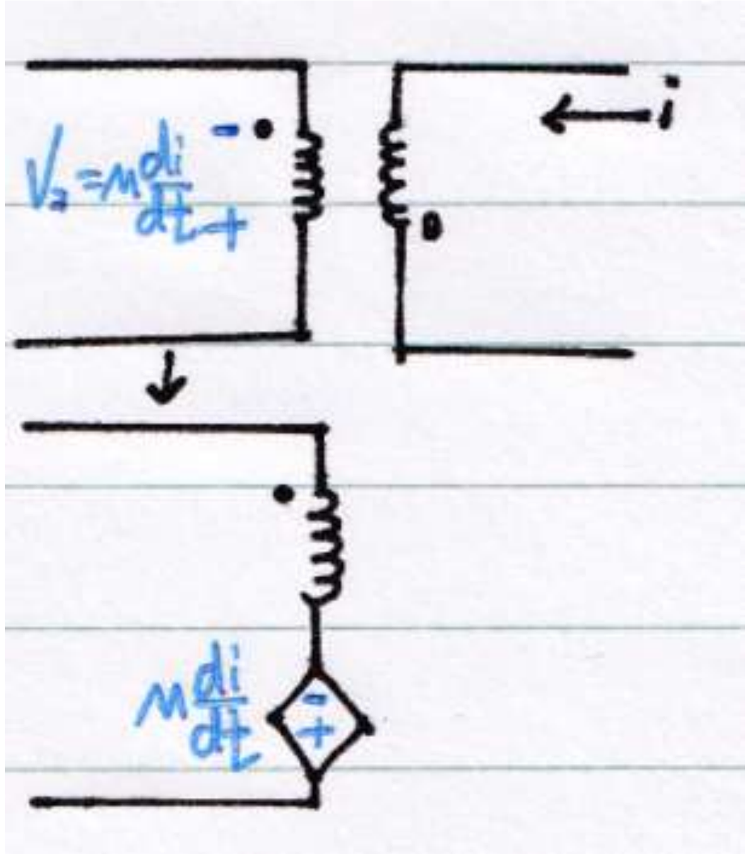
12.2.2 同名端规则

同名端规则：

如果电流进入一个线圈的同名端，则在第二个线圈的同名端处，互感电压的参考极性为正。

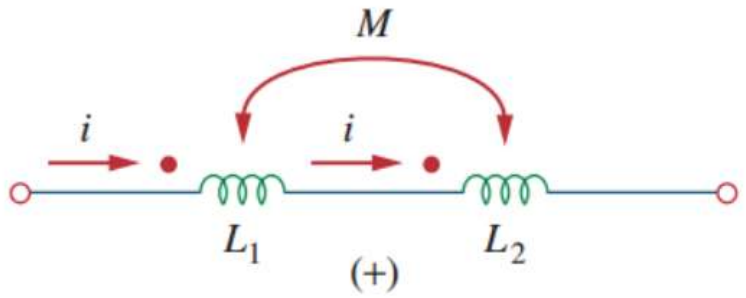
如果电流从一个线圈的同名端流出，则在第二个线圈的同名端处，互感电压的参数极性为负。





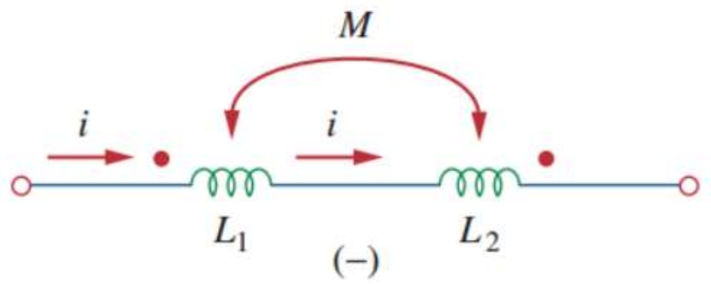
12.3.1 等效耦合电路(Equivalent of coupled inductors)

– Series aiding connection:



$$\boxed{L = L_1 + L_2 + 2M}$$

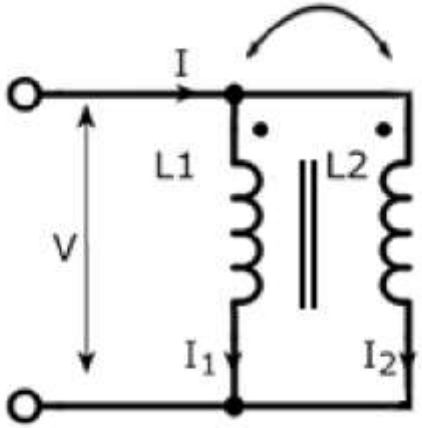
– Series opposing connection:



$$\boxed{L = L_1 + L_2 - 2M}$$

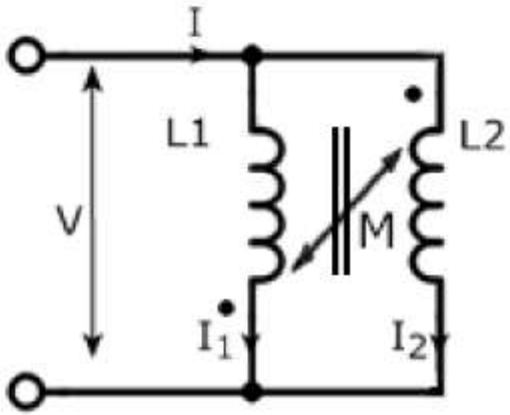
12.3.1 等效耦合电路

同侧并联：



$$\Rightarrow i = \frac{(L_1L_2 - M^2)}{L_1 + L_2 - 2M}\frac{di}{dt} \quad \Rightarrow L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M} \geq 0$$

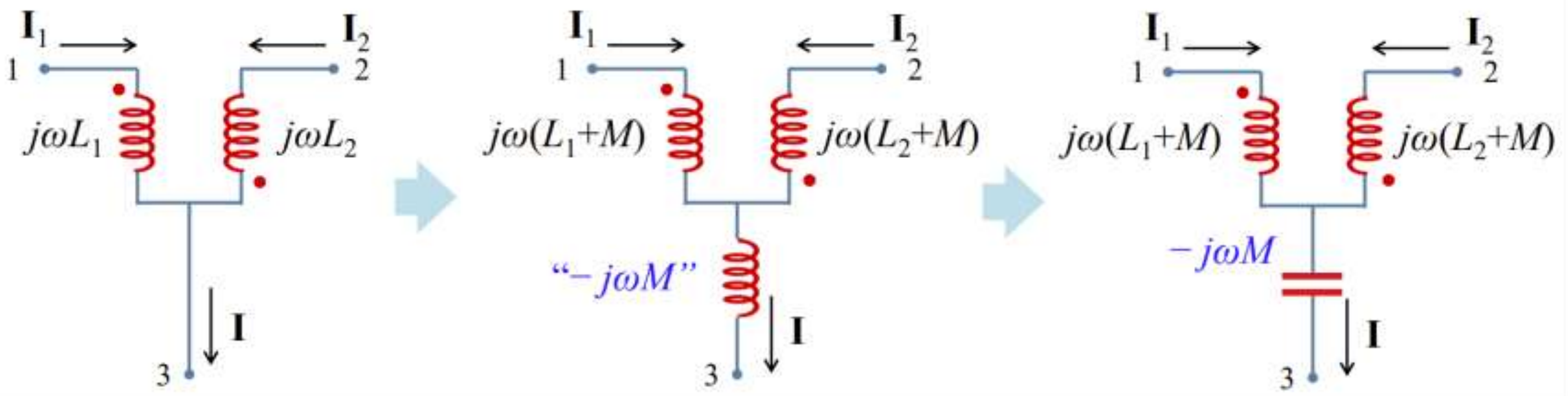
异侧并联：



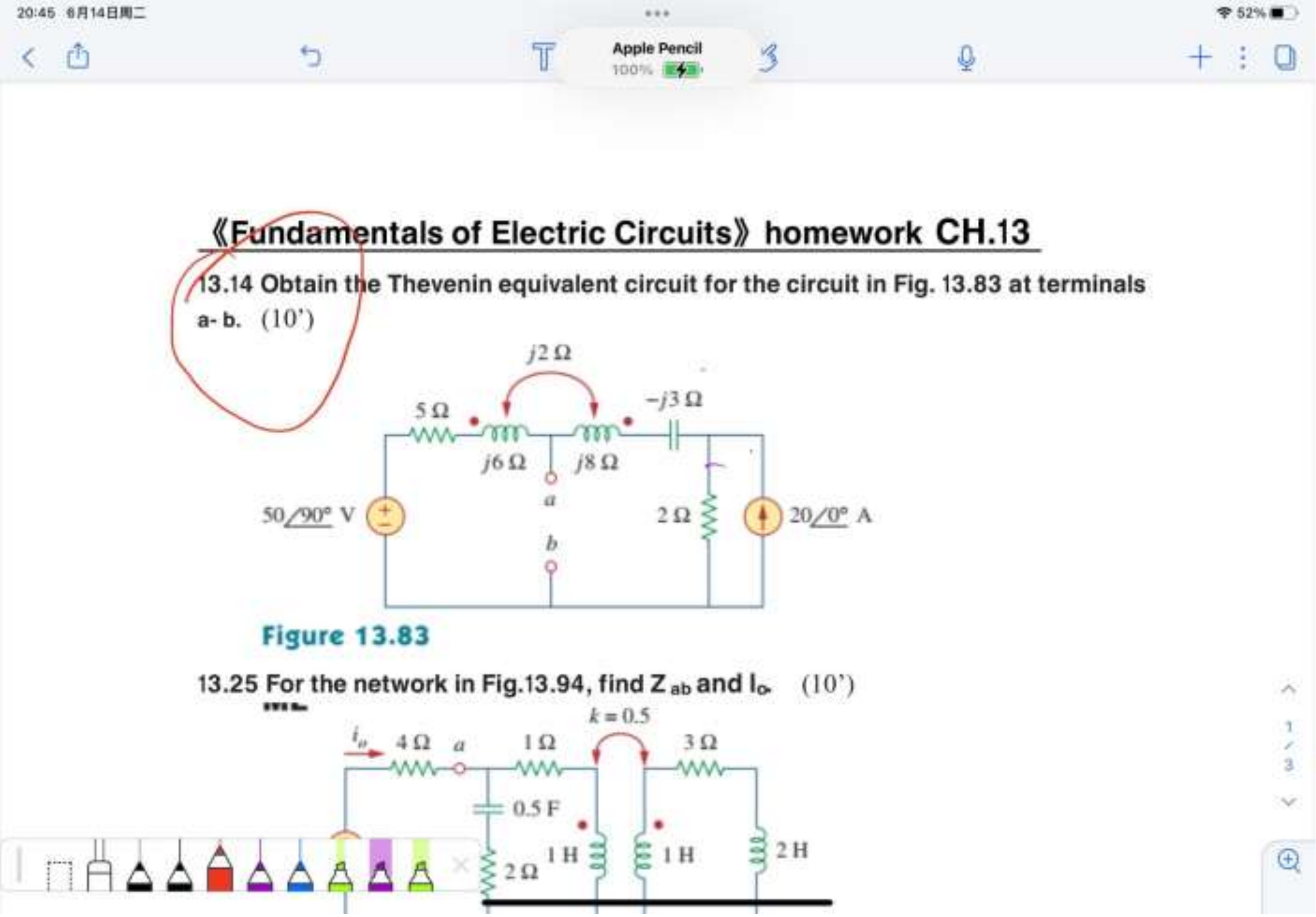
$$\Rightarrow i = \frac{(L_1L_2 - M^2)}{L_1 + L_2 + 2M}\frac{di}{dt} \quad \Rightarrow L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M} \geq 0$$

12.3.1 等效耦合电路

- Parallel-opposing connection (异侧并联):



12.3.1 等效耦合电路



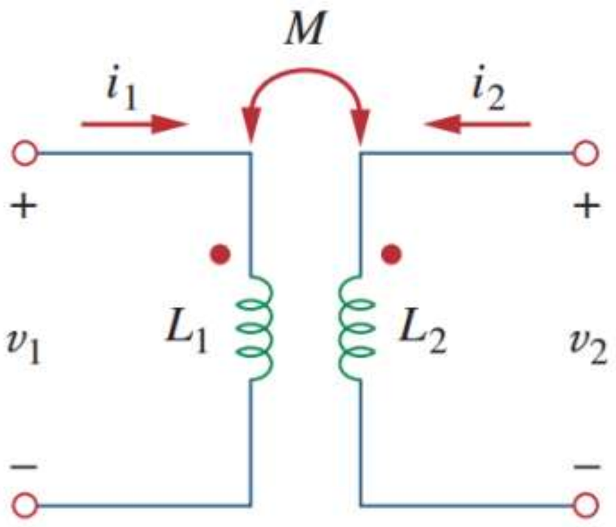
《Fundamentals of Electric Circuits》homework CH.13

13.14 Obtain the Thevenin equivalent circuit for the circuit in Fig. 13.83 at terminals a- b. (10')

Figure 13.83

13.25 For the network in Fig.13.94, find Z_{ab} and I_o. (10')

12.3.2 耦合电路中的能量



$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$

当两个电流均从线圈的同名端流入或流出时，上式互感项取正号，否则负号

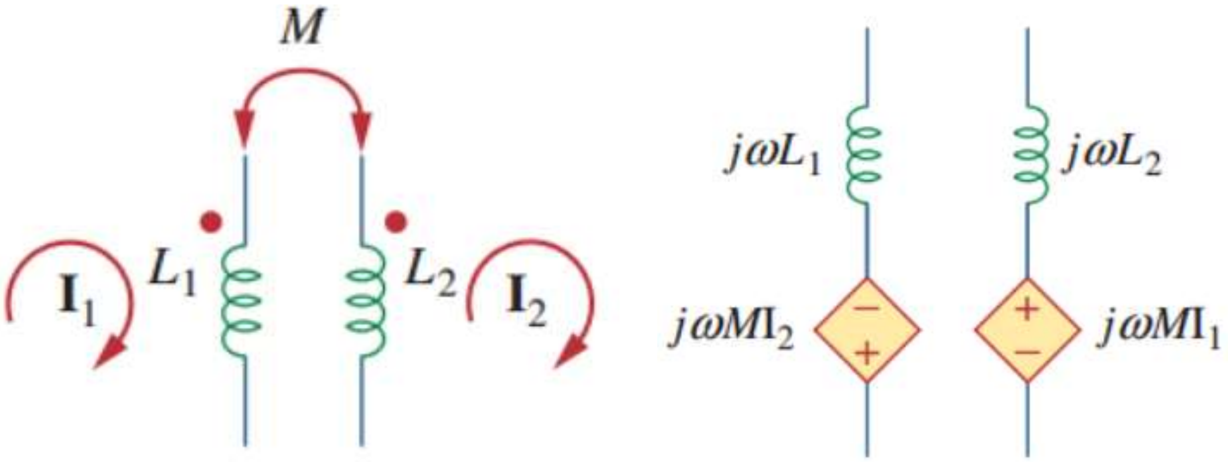
互感M上限 $M \le \sqrt{L_1 L_2}$

Coefficient of coupling 耦合系数 k
两个线圈之间磁耦合程度的一种量度

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

12.3.3等效耦合电路(Equivalent connection)

互感耦合电路模型：受控电压源



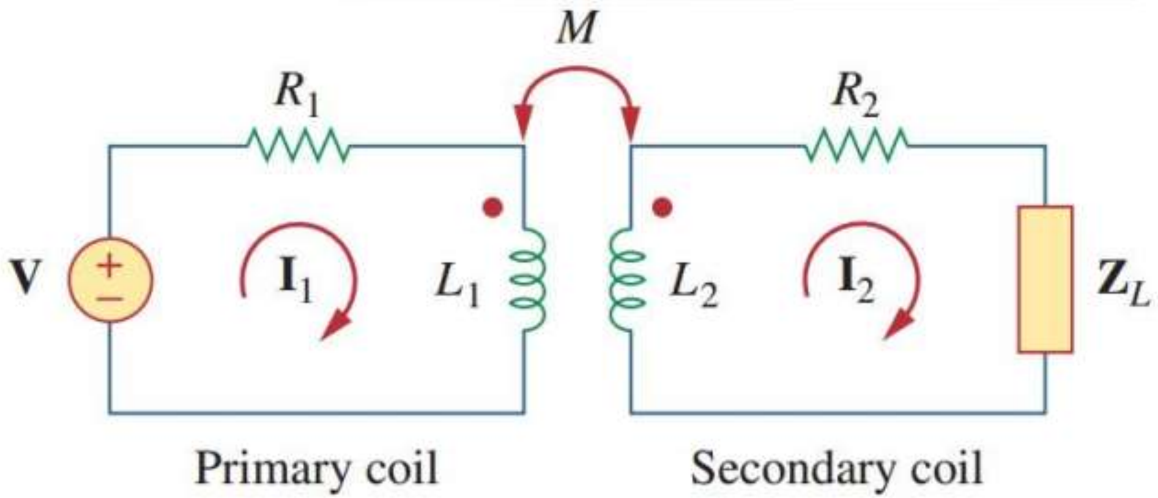
I一定是流过该电感的电流

分析互感耦合电路：
1.利用互感耦合模型，将电路转化为普通电路
2.用KCL求解

12.4 线性变压器(Linear Transformor)

Primary winding: 一次绕组

Secondary winding: 二次绕组



$$\mathbf{V} = (R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + \mathbf{Z}_L)\mathbf{I}_2$$

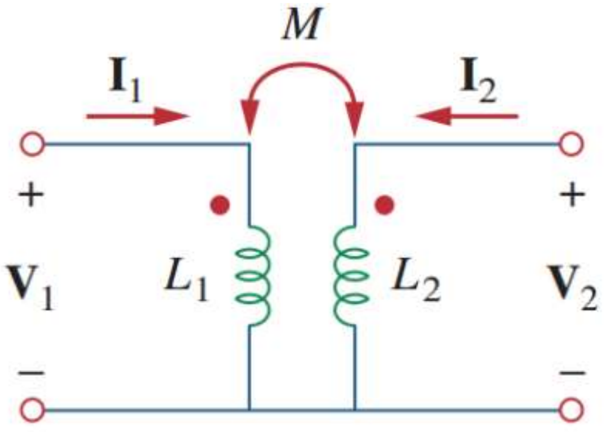
Input impedance 输入阻抗：

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}}{\mathbf{I}_1} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}$$

Reflected impedance 反射阻抗：

$$\mathbf{Z}_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}$$

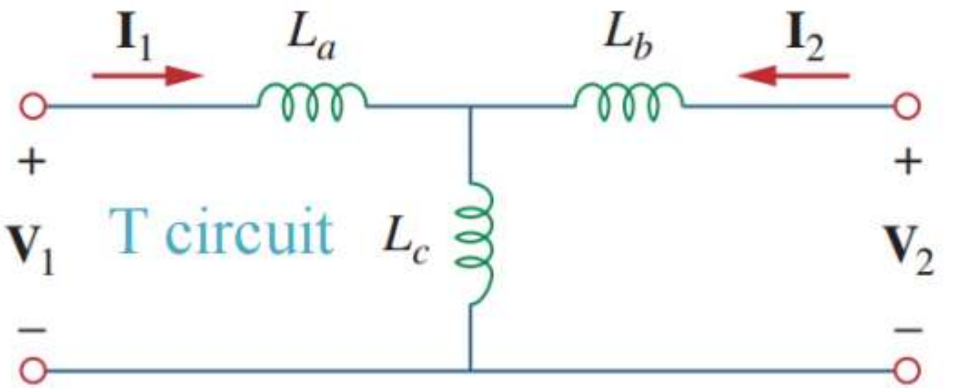
12.4 线性变压器 等效电路



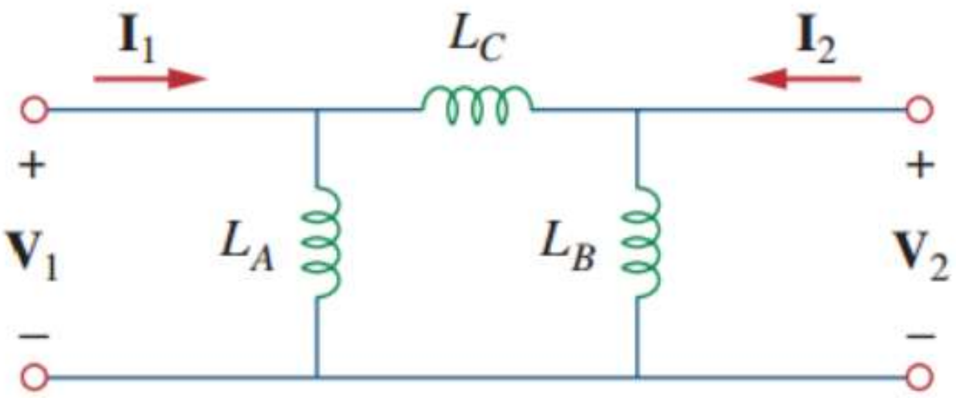
$$\mathbf{V}_1 = j\omega L_1\mathbf{I}_1 + j\omega M\mathbf{I}_2$$
$$\mathbf{V}_2 = j\omega M\mathbf{I}_1 + j\omega L_2\mathbf{I}_2$$

等效后无需考虑各电感磁耦合

注意正负，按照这个图中同名端，只要有一个变反，就取-M；两个变反，取M



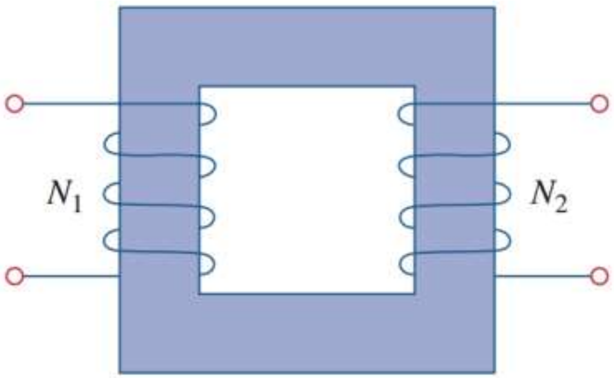
$$L_a = L_1 - M,$$
$$L_b = L_2 - M,$$
$$L_c = M$$

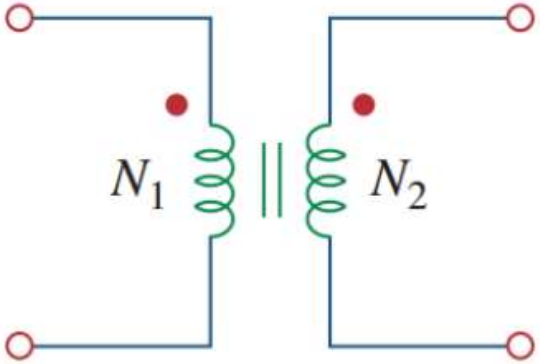


Π circuit

$$L_A = \frac{L_1L_2 - M^2}{L_2 - M},$$
$$L_B = \frac{L_1L_2 - M^2}{L_1 - M}$$
$$L_C = \frac{L_1L_2 - M^2}{M}$$

12.5 理想变压器 (Ideal transformation)





$$\mathbf{V}_1 = j\omega L_1\mathbf{I}_1 + j\omega M\mathbf{I}_2$$
$$\mathbf{V}_2 = j\omega M\mathbf{I}_1 + j\omega L_2\mathbf{I}_2$$

耦合系数k=1 不用考虑磁耦合
电抗很大
一次绕组和二次绕组无损耗

$$\mathbf{V}_2 = j\omega L_2\mathbf{I}_2 + \frac{M\mathbf{V}_1}{L_1} - \frac{j\omega M^2\mathbf{I}_2}{L_1}$$

But $M = \sqrt{L_1L_2}$ for perfect coupling ($k = 1$). Hence,

$$\mathbf{V}_2 = j\omega L_2\mathbf{I}_2 + \frac{\sqrt{L_1L_2}\mathbf{V}_1}{L_1} - \frac{j\omega L_1L_2\mathbf{I}_2}{L_1} = \sqrt{\frac{L_2}{L_1}}\mathbf{V}_1 = n\mathbf{V}_1$$

n 匝数比

12.5 理想变压器

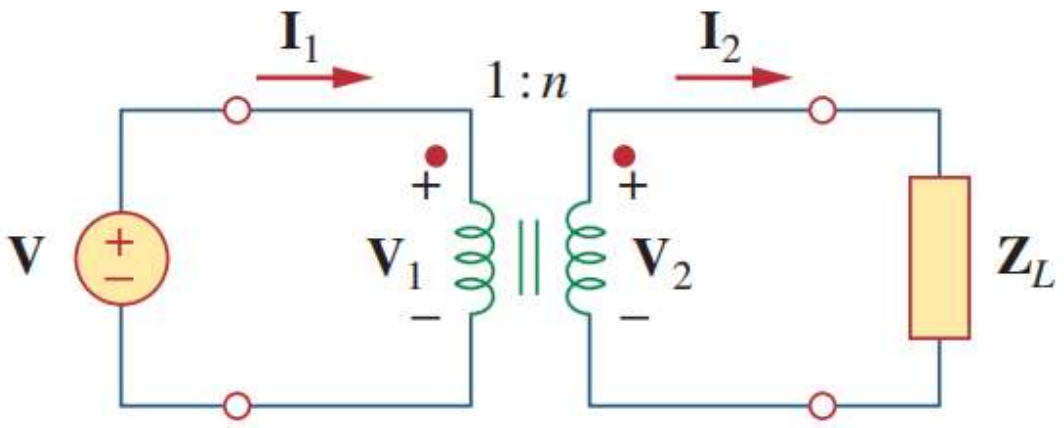
$$v_1 = N_1 \frac{d\phi}{dt}$$

$$v_2 = N_2 \frac{d\phi}{dt}$$

$$\frac{v_2}{v_1} = \frac{N_2}{N_1} = n$$

$$v_1 i_1 = v_2 i_2$$

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{\mathbf{V}_2}{\mathbf{V}_1} = n$$



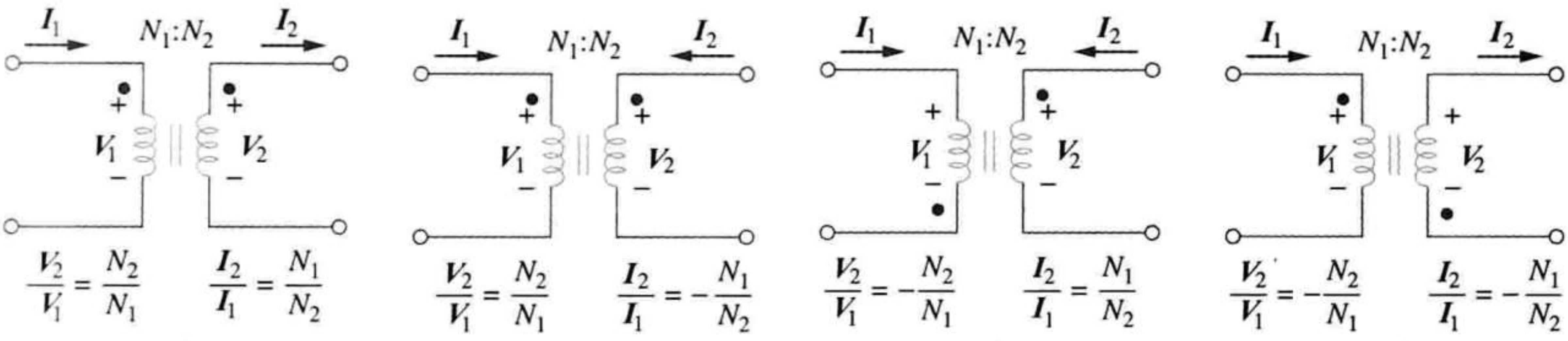
$n > 1$, step–up transformer (升压变压器)

$n = 1$, isolation transformer (隔离变压器)

$n < 1$, step-down transformer (降压变压器)

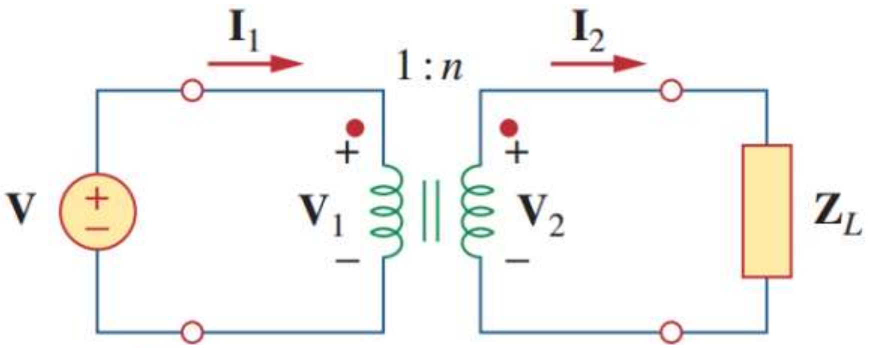
12.5 理想变压器

若同名端处V_1与V_2均为正，或者均为负，采用+n，否则-n
若I_1与I_2均进入或者流出同名端，采用-n，否则+n



a）　b）　c）　d）

12.5 理想变压器 映射

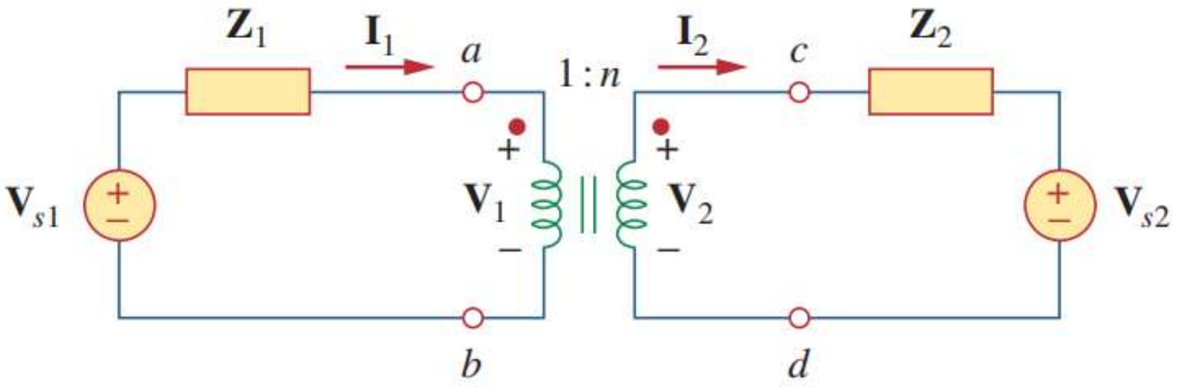


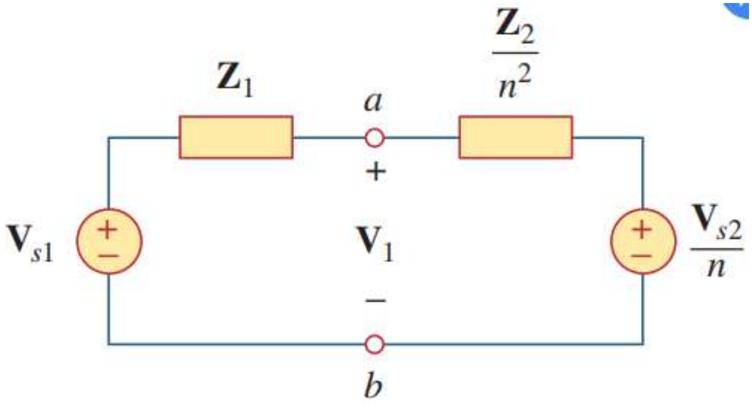
$$\mathbf{V}_1 = \frac{\mathbf{V}_2}{n} \quad \text{or} \quad \mathbf{V}_2 = n\mathbf{V}_1$$

$$\mathbf{I}_1 = n\mathbf{I}_2 \quad \text{or} \quad \mathbf{I}_2 = \frac{\mathbf{I}_1}{n}$$

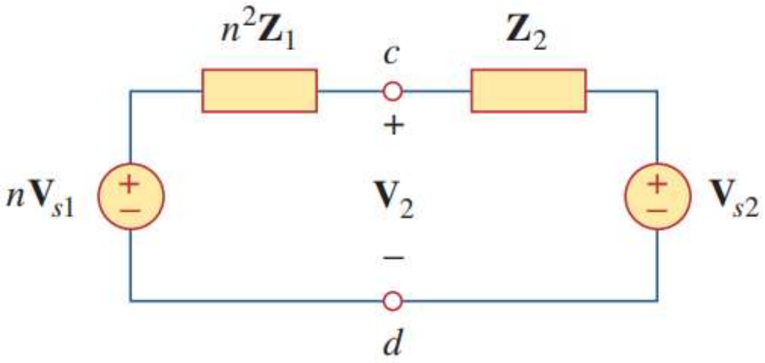
$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{1}{n^2}\frac{\mathbf{V}_2}{\mathbf{I}_2}$$

$$\boxed{\mathbf{Z}_{\text{in}} = \frac{\mathbf{Z}_L}{n^2}}$$





将二次电路映射到一次侧：二次阻抗除以n^2，二次电压除以n，二次电流乘以n



将一次电路映射到二次侧：一次阻抗乘以n^2，一次电压乘以n，一次电流除以n

映射方法只适用于一次二次绕组之间无外部连接的情况，否则只能用网孔分析法

练习1

教材例13.2

Calculate the mesh currents in the circuit of Fig. 13.11.

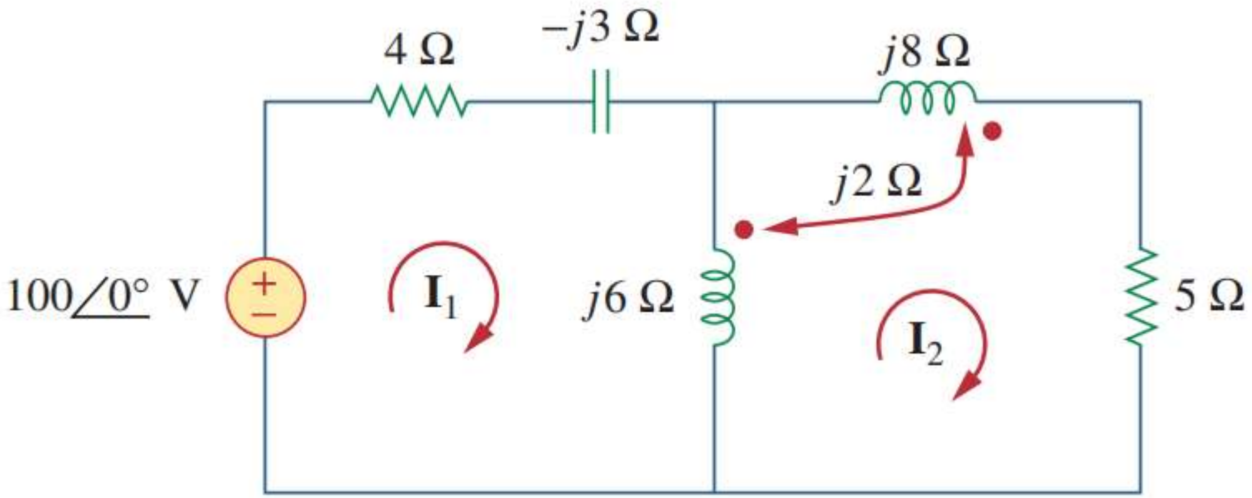


Figure 13.11
For Example 13.2.

练习2

Example 13.3

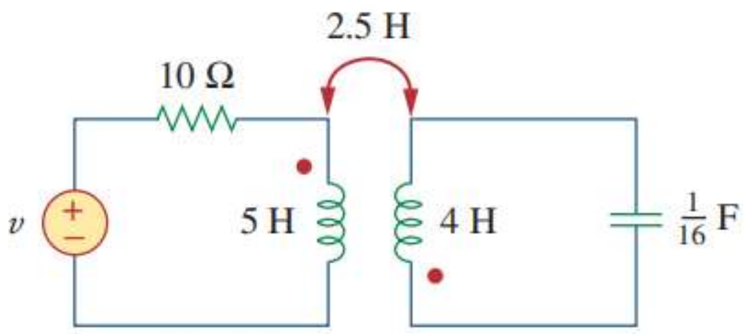


Figure 13.16

Consider the circuit in Fig. 13.16. Determine the coupling coefficient. Calculate the energy stored in the coupled inductors at time $t = 1$ s if $v = 60\cos(4t + 30^\circ)$ V.

Solution:

The coupling coefficient is

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$

练习3

Example 13.6

Solve for $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{V}_o$ in Fig. 13.27 (the same circuit as for Practice Prob. 13.1) using the T-equivalent circuit for the linear transformer.

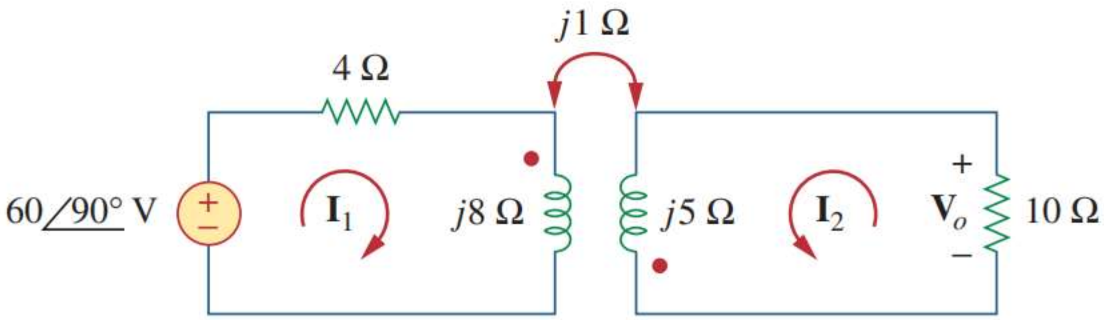
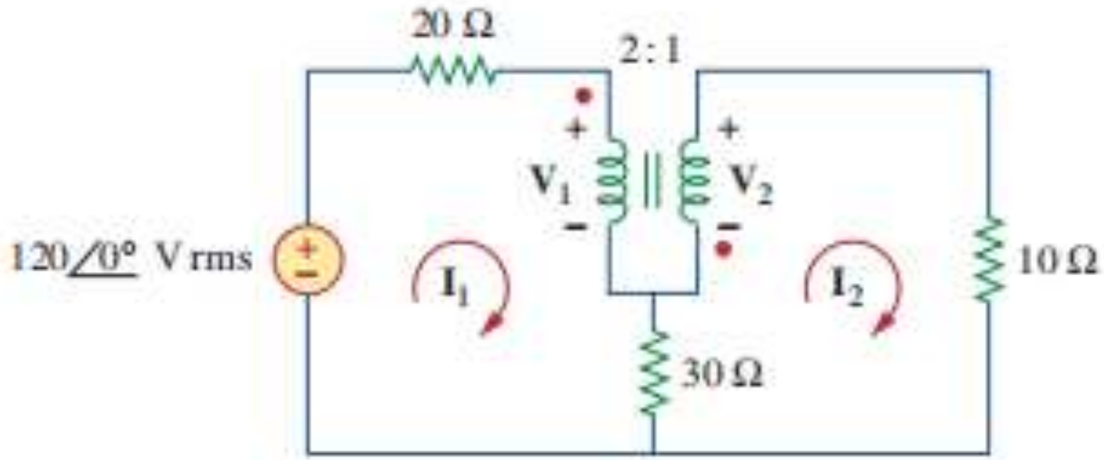


Figure 13.27
For Example 13.6.

练习4

Calculate the power supplied to the 10-Ω resistor in the ideal transformer circuit of Fig. 13.39.

Example 13.9



拉普拉斯变换

13-14

Chapter

13.1 拉普拉斯变换的定义，性质和变换对

13.2 拉普拉斯逆变换

14. 拉普拉斯变换在电路中的应用

13.1 拉普拉斯变换的定义

拉普拉斯变换：将函数f(t)从时域变换到复频域(s域)，变换结果为F(s)，可以极大地简化运算，在处理冲激响应、初值问题的时候尤为有力

Given a function $f(t)$, its Laplace transform, denoted by $F(s)$ or $\mathcal{L}[f(t)]$, is defined by

$$\boxed{\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}\,dt} \tag{15.1}$$

where s is a complex variable given by

$$s = \sigma + j\omega \tag{15.2}$$

One-sided Laplace transform
(Unilateral Laplace transform)

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}\,dt$$

Two-sided Laplace transform
(Bilateral Laplace transform)

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt$$

13.1拉普拉斯变换性质和变换对

表 15-2 拉普拉斯变换对①

$f(s)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
te^{-at}	$\frac{1}{(s+a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t+\theta)$	$\frac{s\sin\theta+\omega\cos\theta}{s^2+\omega^2}$
$\cos(\omega t+\theta)$	$\frac{s\cos\theta+\omega\sin\theta}{s^2+\omega^2}$
$e^{-at}\sin\omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos\omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$

①$t\geqslant 0$；$t<0$ 时，$f(t)=0$。

表 15-1 拉普拉斯变换的性质

性质	$f(t)$	$F(S)$
线性性质	$a_1 f_1(t)+a_2 f_2(t)$	$a_1 F_1(s)+a_2 F_2(s)$
尺度变换性质	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
时域平移性质	$f(t-a)u(t-a)$	$e^{-as}F(s)$
频域平移性质	$e^{-at}f(t)$	$F(s+a)$
时域微分性质	$\frac{df}{dt}$	$sF(s)-f(0^-)$
	$\frac{d^2 f}{dt^2}$	$s^2F(s)-sf(0^-)-f'(0^-)$
	$\frac{d^3 f}{dt^3}$	$s^3F(s)-s^2f(0^-)-$ $sf'(0^-)-\cdots-f^n(0^-)$
	$\frac{d^n f}{dt^n}$	$s^nF(s)-s^{n-1}f(0^-)-$ $sf'(0^-)-\cdots-f^{n-1}(0^-)$
时域积分性质	$\int_0^t f(x)dx$	$\frac{1}{s}F(s)$
频域微分性质	$tf(t)$	$-\frac{d}{ds}F(s)$
频域积分性质	$\frac{f(t)}{t}$	$\int_s^\infty F(s)ds$
时域周期性质	$f(t)=f(t+nT)$	$\frac{F_1(s)}{1-e^{-sT}}$
初值定理	$f(0)$	$\lim\limits_{s\to\infty} sF(s)$
终值定理	$f(\infty)$	$\lim\limits_{s\to\infty} sF(s)$
卷积性质	$f_1(t)*f_2(t)$	$F_1(s)F_2(s)$

13.2拉普拉斯反变换

Step 1: decompose $F(s)$ into simple terms using partial fraction expansion. (部分分式展开)

→ Step 2: find the inverse of each term by matching entries in Table 15.2 (查表得出时域信号)

Possible forms	Methods

$$F(s)=\frac{N(s)}{D(s)}=\frac{a_n s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}$$

1. When $m > n$, $F(s)$ is a proper rational function (有理真分式)

The roots of $D(s)$ are or include:

- Distinct real roots (独立实根)
- Repeated real roots (实重根)
- Distinct complex roots (不等复根)
- Repeat complex roots (多重复根)

2. When $n \leq m$, $F(s)$ is a proper rational function (有理假分式)

Methods

部分分式分解方法

Residue method (留数法)

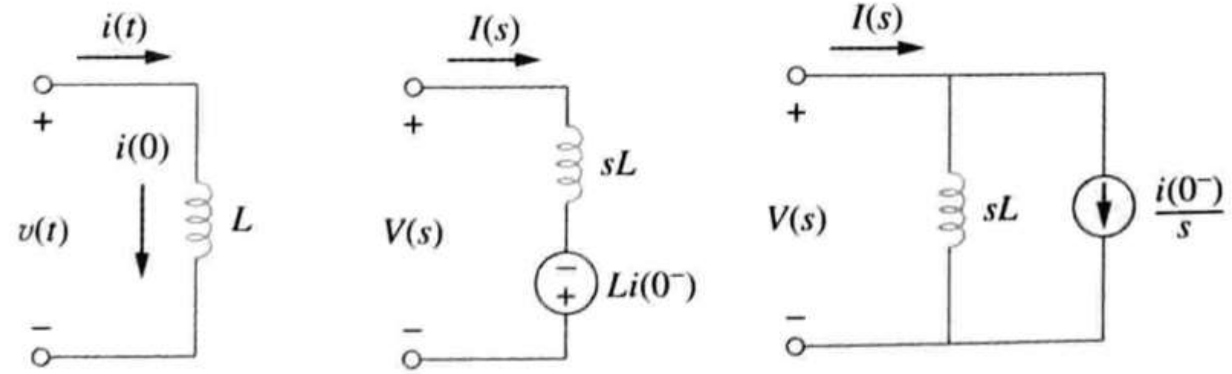
Algebraic method (代数法)

利用拉普拉斯变换求解微分方程

14. 拉普拉斯变换在电路分析中的应用

电阻 $$\boxed{V(s) = RI(s)}$$

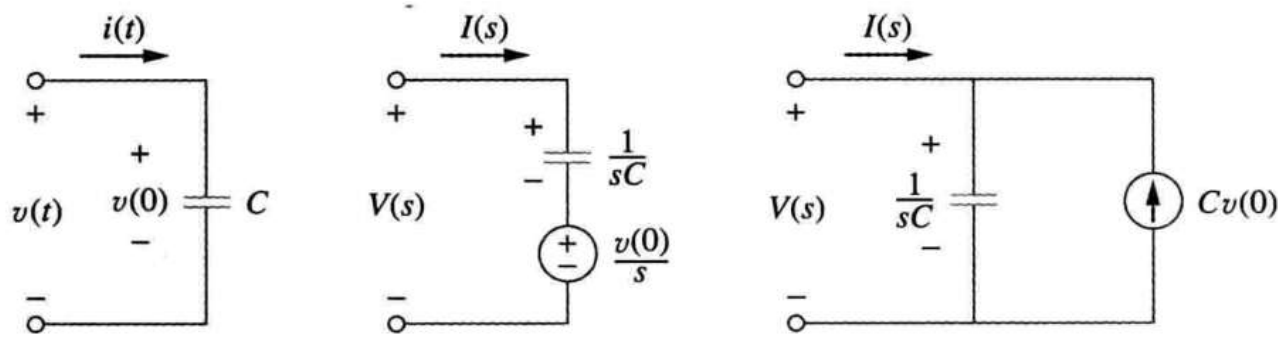
电感 $$\boxed{I(s) = \frac{1}{sL}V(s) + \frac{i(0^-)}{s}}$$



a）时域电路　b）s 域等效电路　c）s 域等效电路

图 16-1　电感器的电路模型

电容 $$\boxed{V(s) = \frac{1}{sC}I(s) + \frac{v(0^-)}{s}}$$



a）时域模型　b）s 域等效模型　c）s 域等效模型

图 16-2　电容器电路模型

14. 拉普拉斯变换在电路分析中的应用

如果假设电感和电容的初始条件为零，则上面的方程化简为：

$$\text{电阻}: V(s) = RI(s), \quad \text{电感}: V(s) = sLI(s), \quad \text{电容}: V(s) = \frac{1}{sC}I(s) \tag{16.9}$$

s 域等效电路如图 16-3 所示。

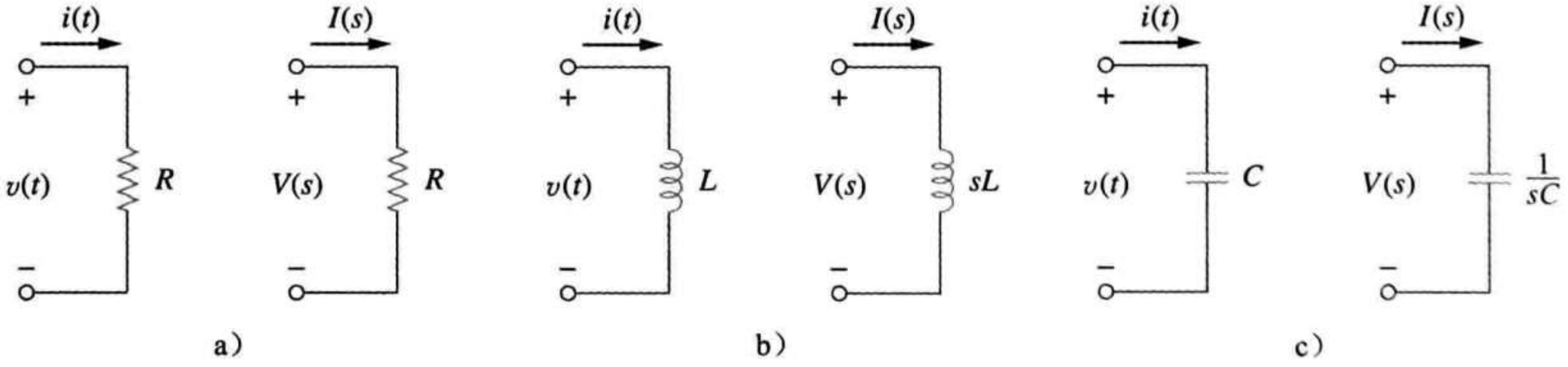


图 16-3　零初始条件下无源元件的时域和 s 域模型

14.1 拉普拉斯变换在电路分析中的应用

1.计算电容两端电压，流过电感电流的初始条件
2.用初始条件将电感、电容转化到s域，将电阻转换到s域
3.将电源转换到s域，方法为求电源时域表达式的拉普拉斯变换
4.运用电路分析方法解电路
5.将结果拉普拉斯逆变换转换到时域

电感、电容变换到s域所增加的电源不能反变换到时域
利用电路分析方法解题：叠加定理，戴维南.....

THANKS